

RESEARCH METHODOLOGIES AND EMERGING TECHNOLOGIES

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Assignment 2- Contextual Report

“RentHub – A Digital Home Rent and Tenant Management Platform”



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Abstract

Rental housing market in the urban centers of Nepal and in the Kathmandu Valley especially is in a very informal state and is highly dependent on brokers, there is no verified property data, and the landlords are inefficient in maintaining records manually. This report presents RentHub, a mobile-first responsive centralized web application that intends to rationalize the rental procedure by means of a role-based management platform that is based on the PERN stack (PostgreSQL, Express.js, React.js, and Node.js).

One of the main technical innovations of the suggested system is the implementation of a Geospatial Exponential Search algorithm. This process overcomes the problem of empty result that is inherent to the conventional platforms through the gradual and logarithmic expansion of the search area around the user-selected landmarks in case no immediate matches are found. In addition, architecture has a Deterministic Rule-Based Chatbot to ensure that property search is made easier with the help of guided conversational interface so that the data is accurate in the field where financial and legal information matters.

The following report captures the overall research and design stage of the project. Core problem statement is statistically well-supported through the primary research that involved more than 50 participants (tenants and landlords). The statistics indicate that there is almost a unanimous request for bidirectional identity verification and landmark-based search features. Most importantly, the data proves that about 70 percent of users think that such a platform can significantly decrease the dependence on a broker and enhance market transparency. The proposed design complies with the Individual Privacy Act 2075 that provides ethical governance of data. RentHub, by connecting theoretical algorithmic ideas with real-world software engineering, offers a proven architecture design of a transparent, efficient, and scalable rental marketplace in Nepal.

Acknowledgement

I would also like to acknowledge all the individuals who have helped to make this project successful in its research and design phase.

To begin with, I would like to say that I would not have reached this point without the invaluable guidance, constant encouragement, and insightful feedback of my supervisor, Mr. Pawan KC, during this semester. His technical skills and scholarly assistance have played a key role in developing the architectural principles of RentHub and perfecting the research procedure.

I also owe my debt to the University of Bedfordshire and its faculty that enabled me to have the academic resources and environment to carry out this project. The technical issues of the current study have been met by the knowledge gained during the BSc (Hons) Software Engineering program.

The 50+ participants, landlords and tenants within the Kathmandu Valley have been given a special word of thanks as they took part in my primary research survey. Their candid feedback and statistics gave them the empirical evidence they needed to confirm the objectives and marketability of the project.

Table of Contents

1. Introduction	1
1.1. Project Background	1
1.2. Problem Statement	1
1.3. Proposed Solution: The RentHub Digital Ecosystem	2
1.4. Aims and Objectives	3
1.4.1. Project Aim	3
1.4.2. Specific Objectives	3
1.5. Project Scope	4
1.5.1. Functional Scope (In-Scope)	4
1.5.2. Technical Constraints (Out-of-Scope)	4
1.6. Intellectual Challenges: Adapting Exponential Search for Geospatial Use	5
1.6.1. The Logic Shift: From Arrays to Earth Coordinates	5
1.6.2. Technical Engineering Hurdles	5
1.7. Rationale and Social Impact	6
1.7.1. Economic Empowerment	7
1.7.2. Conformance with National Digital Policy	7
1.7.3. Supporting Urban Mobility	8
1.8. Structure of the Report	8
2. Contextual Review (Literature Review)	9
2.1. Overview of Research Themes	9
2.2. Review of Existing Products	9
2.2.1. Analysis of Local Competitors (HamroBazar, MeroProperty)	9
2.2.2. Comparative Analysis of Global Models (Zillow, Airbnb)	9
2.3. Thematic Review of Technical Foundations	10
➤ Theme 1: Advanced Spatial Algorithms & R-Tree Indexing	10
1. The former is Exponential Search Logic in Unlimited Domains (Bentley and Yao, 1976).	10
2. R-Tree Hierarchical Spatial Indexing (Guttman, 1984)	10
3. PostGIS Spatial Joins Calculations in Computational Geometry (IEEE Xplore, 2025)	11
4. Exponential Search Logic (BMR Education, 2026)	12
➤ Theme 2: Deterministic Conversational AI vs. Probabilistic Models	12

5.	Deterministic vs. Probabilistic Chatbot Frameworks (HCI Research)	12
6.	Predictive Lead Scoring & Qualification (Chaki and Dey, 2020)	13
7.	Intent Recognition and Natural Language Understanding (NLU) (Gao et al., 2025)	13
➤	Theme 3: System Reliability: ACID Compliance in the PERN Stack	14
8.	MERN vs. PERN Stack: The ACID Compliance Factor (CodeWithAntonio, 2025)	14
9.	Effect of Test-Driven Development (TDD) on the Stability of a System (Benardo, 2024)	15
10.	Scalability database Connection Pooling (Parnin, 2024)	15
11.	Test-Driven Development (TDD) Lifecycle (Fireship, 2024)	16
➤	Theme 4: Security Architecture (JWT, RBAC, and SQL Injection Prevention)	17
12.	JSON Web Tokens Stateless Authentication (Node.js Foundation, 2025)	17
13.	Role-Based Access Control (RBAC) Hierarchies (Vercel, 2026)	17
14.	SQL Injection (W3C Security, 2025) Defensive Programming.	18
15.	API Shielding (O'Reilly, 2024) and Cross-Origin Resource Sharing (CORS) (O'Reilly, 2024).	19
➤	Theme 5: Legal Context: The Nepal Individual Privacy Act 2075	19
16.	Sovereignty of Data and Foundations of the Constitution (Article 28, Constitution of Nepal)	19
17.	Purpose Limitation and Informed Consent (Section 21, Privacy Act 2075)	20
18.	Right to Correction and Data Accuracy (Section 23, Privacy Act 2075)	21
19.	Obligations of Data Security and Liability of Breach (Privacy Act 2075)	21
➤	Theme 6: Advanced Implementation (UI/UX and System Performance)	22
20.	Mobile First Responsive Design (Zurb/Bootstrap Standards, 2025)	22
21.	Live Chats WebSockets (Socket.io Documentation, 2026)	23
22.	Skeleton Screens and Perceived Performance (Nielsen Norman Group, 2025)	23
23.	Server-side Image Compression (Sharp Documentation, 2026)	24
➤	Theme 7: Algorithmic Matching & Data-Driven User Retention	24
24.	The BANT Framework of Automated Lead Qualification (Molin AI, 2026)	25
25.	Feedback Loops and Habit-Forming UX (Hook Model, 2025)	25
26.	Data Normalization of Multi-Criteria Filtering (Date and Pratt, 2024)	26
2.4.	Summary of Research Gaps	26
2.4.1.	The first issue is the Algorithmic Discovery Gap (Spatial Strategy)	26
2.4.2.	Trust and "Lead Fatigue" Gap (Conversational Strategy)	27

2.4.3.	Data Integrity and Reliability Gap (Architectural Strategy).	27
2.4.4.	Legal and Privacy Gap (Compliance Strategy).	27
2.5.	Conclusion and Technical Justification for Chosen Tools.	27
2.5.1.	The PERN Stack (PostgreSQL, Express, React, Node.js)	27
2.5.2.	Deterministic AI and Algorithmic search	28
2.5.3.	Security & Legal Compliance.	28
2.6.	Final Summary	28
3.	Primary Research & Requirements Validation	29
3.1.	Research Methodology and Ethical Considerations	29
3.1.1.	Research Methodology: Mixed-Methods Approach.	29
3.1.2.	Ethical Considerations	29
3.2.	Quantitative Analysis and Findings	30
3.2.1.	Kathmandu Valley Demographics.	30
3.2.2.	Pivot Analysis: Search Behavior and Spatial Logic.	31
3.2.3.	Measures of Security, Trust and Identity Verification.	32
3.2.4.	Demand: Management and Automation.	33
3.3.	Qualitative Insights: User Pain Points and Feature Desires.	34
3.3.1.	Thematic Analysis of User Feedback.	35
➤	Theme A: The Trust and Transparency Gap.	35
➤	Theme B: Disputes in Post-Rental Management.	35
➤	Theme C: Market Inefficiencies (The "Broker" Problem)	35
3.3.2.	Evidence of Qualitative Feedback	36
3.3.3.	Feature Desire Mapping	38
3.4.	Validation: How Research Data Informed System Design	38
3.4.1.	Mapping User Pain points to Technical Solutions.	38
3.4.2.	Indication of Informed UI/UX Design.	39
3.4.3.	Final Validation Statement.	40
4.	Project Plan & Methodology	40
4.1.	Software Development Life Cycle (SDLC): The Agile Scrum Approach	40
4.1.1.	Why Agile for a Solo Project?	41
4.1.2.	The Scrum Framework Adaptation	41
4.1.3.	The Definition of Done (DoD).	42

4.1.4.	Development Workflow (Solo-Pipeline).....	42
4.2.	Work Breakdown Structure (WBS) for Research and Design.....	42
4.3.	Project Timeline.....	43
4.3.1.	Phase 1: Research and Design	44
4.3.2.	Phase 2: Implementation Roadmap.....	45
4.4.	Risk Analysis and Mitigation Matrix	47
4.5.	Risk Assessment Visualization	49
4.6.	Resource Requirements (Software, Hardware, and Simulation Tools)	49
4.6.1.	Hardware Requirements.....	50
4.6.2.	Software Requirements	50
4.6.3.	Architectural & Simulation Tools.....	51
5.	Artefact Planning (Technical Design)	51
5.1.	System Requirements Specification.....	51
5.1.1.	Functional Requirements (MoSCoW Prioritization).....	51
5.1.2.	Non-Functional Requirements (Performance, Security, Privacy).....	53
5.2.	Process and Information Architecture	53
5.2.1.	Use Case Analysis.....	53
5.2.2.	Data Flow Architecture	54
5.2.3.	System Interaction and Dynamic Behavior.....	55
5.3.	Database and Schema Design.....	56
5.3.1.	Logical Data Model (ERD).....	56
5.3.2.	Software Class Structure	57
5.3.3.	Geospatial Data Storage (PostGIS Integration)	58
5.4.	Logic and Algorithm Design.....	58
5.4.1.	Recursive Radius Expansion (Exponential Search Logic Flow).....	58
5.4.2.	Finite State Machine (FSM) for Chatbot Conversations.....	59
6.	Quality Assurance and Testing Plan	60
6.1.	Verification and Validation Strategy.....	61
6.2.	Specifications of the proposed test case.....	61
6.3.	Security & Privacy Assurance (Non-Functional).....	62
6.4.	Research-Based Usability Evaluation	62
7.	Conclusion and Research Synthesis.....	63

7.1.	Final Summary of Findings	63
7.2.	Critical Reflection on Design Challenges	63
7.3.	Implementation Readiness and Future Outlook.....	63
References.....		64
Appendices		66

Table of Figures

Figure 4: - Age Group Chart.....	30
Figure 5: - Filter Preference plot	31
Figure 6: - Radius Expansion Threshold Chart.....	32
Figure 7: - Identity Verification plot.....	32
Figure 8: - Privacy Concerns Plot.....	33
Figure 9:- Information Storage Method Chart.....	34
Figure 10: - Importance of Feature Plot	34
Figure 11: - Additional Comments Section.....	37
Figure 12: - Conversational UI (Chatbot) Chart	40
Figure 1: - Gantt chart for your RentHub Research phase.....	45
Figure 2: - Gantt chart for the RentHub Development Phase.....	47
Figure 3: - Risk Heat Map	49
Figure 13: - Use Case Diagram.....	54
Figure 14: - Data Flow Diagrams(level 0).....	54
Figure 15: - Data Flow Diagrams (level 1).....	55
Figure 16: - Sequence Diagram.....	56
Figure 17: - Entity Relationship Diagram (ERD).....	57
Figure 18: - Class Diagram	58
Figure 19: - Exponential Search Logic Flow.....	59
Figure 20: - Chatbot Conversation Flow	60

Table of Tables

Table 1: - Problem Statement	1
Table 2: - Core Functional Modules and Technical Solutions	2
Table 3: - Specific Objectives	3
Table 4: - Functional Scope (In-Scope).....	4
Table 5: - Algorithmic Comparison.....	6
Table 6: - Social Impact and Value Creation	7
Table 7: - Structure of the Report	8
Table 15: - Feature Desire Mapping.....	38
Table 16: - Mapping User Pain points to Technical Solutions.....	39
Table 8: - lean Solo-Scrum process.....	41
Table 9: - Work Breakdown Structure (WBS).....	43
Table 10: - Phase 1: Research and Design	44
Table 11: - Phase 2: Implementation Roadmap.....	46
Table 12: - Risk Analysis and Mitigation Matrix	48
Table 13: - Hardware Requirements.....	50
Table 14: - Software Requirements.....	50
Table 17: - Functional Requirements (MoSCoW Prioritization).....	52
Table 18: - Non-Functional Requirements.....	53
Table 19: - Specifications of the proposed test case.....	62

1. Introduction

1.1. Project Background

The Nepal real estate and rental industry and especially in Kathmandu Valley is in the midst of an unregulated fast growth. Being the economic and educational center of the country, Kathmandu is the constant center of flow of new students and working population, which generates a high-stress demand for residential rent. In spite of this expansion, the market is still in a conventional, informal structure.

The discovery stage of renting is currently dependent on the physical To-Rent signboards or informal agents called brokers. These brokers usually work unregulated, and they make huge commissions to both parties by giving out uncertified information. Most of the landlords continue to use paper registers or basic mobile notes to track tenancies, which results in loss of data and financial conflicts at the management side. RentHub is imagined as a mobile-first responsive web platform that will fill this digital divide. Using the PERN stack (PostgreSQL, Express.js, React.js, Node.js), the project will turn the rental process, which involves a lot of manual work and high friction, into a transparent and data-driven digital process.

1.2. Problem Statement

The table below summarizes the systemic failures found in the primary research of 50+ respondents:

Table 1: - Problem Statement

ID	Problem Domain	Description of Inefficiency
1.2.1	Discovery	Broker Dependency: Tenants pay high commissions (often 1 month's rent) to unregulated intermediaries to find listings that should be made public.
1.2.2	Management	Manual Record-Keeping: Landlords lack digital tools for lease tracking, rent receipts, and maintenance logs, leading to administrative burdens.

1.2.3	Security/Trust	The "Trust Gap": Prevalence of "ghost listings" and unverified advertisements on social media creates a low-trust environment for digital adoption.
1.2.4	UX/Algorithm	Static Search: Current platforms use rigid boundaries (Wards). If no result is found, users receive an "Empty Result" page with no alternatives.

1.3. Proposed Solution: The RentHub Digital Ecosystem

The suggested solution will be a centralized and mobile-first responsive web app, which will transform the entire rental lifecycle to digital. RentHub offers a formalized setting to property discovery and tenancy management by leaving behind the informal brokerage platform.

Table 2: - Core Functional Modules and Technical Solutions

Module	Feature	Technical Implementation	Value Proposition
Smart Discovery	Exponential Radius Search	Integration of PostGIS with a custom Node.js looping logic that doubles the search radius (e.g., 500m to 1km) if results are null.	Eliminates "Empty Result" frustration and provides intelligent alternatives near landmarks.
User Assistance	Rule-Based Chatbot	A Finite State Machine (FSM) drives conversational interface to collect user preferences (Budget, BHK, Location).	Simplifies the search process for non-technical users and provides automated recommendations.
Tenancy Management	Digital Lease & Payment Logs	Relational database scheme in PostgreSQL to track lease start/end dates and rent payment history.	Replaces manual paper registers with permanent, searchable, and timestamped digital records.
Property Control	Landlord Dashboard	React-based CRUD interface for managing listings, image galleries, and maintenance ticket status.	Empowers landlords to manage multiple properties from a single centralized interface.
Security & Trust	Role-Based Access (RBAC)	JWT (JSON Web Tokens) and Bcrypt password hashing for secure authentication of Admins, Landlords, and Tenants.	Ensures data privacy and builds market trust through verified user roles.

1.4. Aims and Objectives

1.4.1. Project Aim

The main objective of the project is to architect and design RentHub, a unified digital platform that will bring property discovery to the modern era by using intelligent algorithms and offer a structured and safe platform to lease and manage tenants in the Nepalese market.

1.4.2. Specific Objectives

The following table breaks down the project objectives into clear, measurable phases:

Table 3: - Specific Objectives

ID	Objective Phase	Measurable Outcome
O1	Contextual Research	Conduct a comprehensive review of Geospatial Search Algorithms and secure PERN stack architectures, citing at least 15-20 academic sources.
O2	Primary Validation	Analyze data from 50+ survey participants to validate the demand for landmark-based search and direct-to-landlord communication.
O3	Architectural Design	Design a scalable PostgreSQL database schema with PostGIS integration, documented through an Entity Relationship Diagram (ERD).
O4	Algorithmic Logic	Develop a technical blueprint for the Exponential Search radius expansion logic and a Finite State Machine (FSM) for the rule-based chatbot.
O5	UX/UI Planning	Create high-fidelity wireframes and a System Architecture Diagram to guide the frontend development in React.js.
O6	Quality Planning	Establish a Test-Driven Development (TDD) framework and a "Definition of Done" (DoD) to ensure system reliability during implementation.
O7	Project Management	Formulate a 10-week Agile/Scrum Roadmap and GANTT chart to ensure the timely delivery of a functional artefact in Semester 2.

1.5. Project Scope

RentHub is centered around the development of a high-performance, responsive web platform that supports the foundation of the rental lifecycle discovery, management, and communication.

1.5.1. Functional Scope (In-Scope)

The following table details the primary features that will be researched, designed, and implemented:

Table 4: - Functional Scope (In-Scope)

Category	Feature	Description
Authentication	Role-Based Access (RBAC)	Secure login/registration for three distinct roles: Admin, Landlord, and Tenant using JWT.
Property Discovery	Exponential Search	A radius-based search that expands (e.g., 1km, 2km, 4km) around a landmark if zero results are found.
User Assistance	Rule-Based Chatbot	A conversational agent to collect user preferences (budget, room type) and suggest matching properties.
Landlord Tools	Property Management	Full CRUD (Create, Read, Update, Delete) capability for property listings, including image uploads and status updates.
Tenancy Tracking	Lease & Payment Logs	Digital record-keeping for lease start/end dates and a historical log of rent payments.
Communication	Maintenance Requests	A ticketing system where tenants can report issues (e.g., plumbing, electrical) and landlords can update the resolution status.

1.5.2. Technical Constraints (Out-of-Scope)

In order to keep the project viable and to deliver a high-quality core product, the following factors will have no place on the current development road map:

- Native Mobile Applications: The initiative will be a Responsive Web Application (accessible through mobile browsers) and it will not contain any native.apk (Android) or.ipa (iOS) applications.
- Real-Time GPS Tracking: The system uses a combination of fixed coordinates (latitude/longitude) of properties and landmarks instead of the real-time, live user GPS positioning.
- Integrated Payment Gateway: Although the system will record payments to keep the records, in this version, it will not handle live financial transactions with third-party APIs such as eSewa or Khalti.
- Higher AI/NLP: The chatbot will not use advanced Natural Language Processing (NLP) or Large Language Models (LLM) but will use deterministic Rule-Based Logic (Finite State Machine).
- Automated Legal Documentation: The system does not automatically create and digitally sign legally binding government contracts; it merely tracks leases.

1.6. Intellectual Challenges: Adapting Exponential Search for Geospatial Use

The main technical difficulty of this project is mathematical and logical adaptation of the Exponential Search algorithm. Historically, this algorithm has been applied to locate the range within which a target value lies in a sorted array, which is not bound, with a time complexity of $O(\log n)$. In the case of RentHub, this reasoning is converted into a Logarithmic Radius Expansion algorithm to address the "Empty Result" issue in property finding.

1.6.1. The Logic Shift: From Arrays to Earth Coordinates

A typical search involves a user choosing a landmark and the system will search within a predefined circle (e.g., 500m). In case there are no properties, a user is informed about the error No Results Found. The intellectual challenge in this case is to make use of an Elastic Search, imitating human behavior in this case, which is to automatically search far out when the immediate locality is clear.

1.6.2. Technical Engineering Hurdles

The application of this to a stack of PERN (PostgreSQL, Express, React, Node) brings three different challenges:

Spatial Indexing (PostGIS Integration): The map data is too slow to be searched with standard SQL queries that are in the form of linear searches. The difficulty is how to apply PostGIS to execute R-Tree (GIST) Indexing. This would enable the database to disregard 99% of the map and only visualize the properties in the growing circles, which makes the site fast even with thousands of listings.

Recursive Query Logic in Node.js: The backend will have to deal with a recursive loop, which will only be invoked when the first database result is null value. This means that dealing with these asynchronous "chained" requests and maintaining the UI responsive in React takes sophisticated Promises and state management skills.

Setting the Threshold Cap: A geographical search has a Relevance Limit, as opposed to an array, which is theoretically infinite. One significant issue is how to establish a mathematical limit (e.g., $R_{max}=5km$) to ensure that the algorithm does not present results that are too distant to be of practical use to retain the Quality of Result (QoR).

Table 5: - Algorithmic Comparison

Feature	Traditional Array Search	RentHub Geospatial Search
Data Structure	Sorted Linear Array	Latitude/Longitude Coordinates
Search Space	Array Indices ()	Radius () in Meters
Growth Pattern	$i = i * 2$	$r = r * 2^n$ (500m, 1km, 2km...)
Stopping Condition	Target found / End of array	Property found / 5km Threshold

By successfully adapting this algorithm, the project demonstrates **Algorithmic Competency**. It moves beyond basic "If-Else" filtering to provide an "Intelligent Search" experience specifically tailored for the dense, landmark-heavy urban layout of the Kathmandu Valley.

1.7. Rationale and Social Impact

The idea behind coming up with RentHub lies in the fact that the Nepalese rental market is experiencing a digital divide. House hunting is a manual and high friction process in comparison to other sectors such as the banking industry and food delivery which have been digitized.

Table 6: - Social Impact and Value Creation

Impact Area	Current Situation (Manual)	RentHub Impact (Digital)	Social Value
Financial	High broker commissions (often 50%–100% of one month's rent).	Direct-to-landlord connection via a verified platform.	Cost Savings: Especially for students and low-income workers.
Transparency	Verbal agreements and paper receipts lead to frequent security deposit disputes.	Immutable digital payment logs and lease tracking.	Legal Clarity: Reduces conflict between tenants and landlords.
Efficiency	Physical "To-Rent" sign hunting is time-consuming and exhausting.	Geospatial Exponential Search finds the nearest options instantly.	Time Optimization: Streamlines urban migration for professionals.
Trust	High prevalence of "ghost listings" and fake social media posts.	Role-based verification and landmark-centric data.	Market Integrity: Builds a safer digital ecosystem for all users.

1.7.1. Economic Empowerment

RentHub returns the economic value to the users by eliminating unregulated intermediaries (brokers). To a student in Kathmandu, the amount saved as a brokerage fee can be close to a month of living. This initiative makes housing information more democratic so that the capacity of an individual to locate housing is not based on his/her social network and capacity to pay an intermediary.

1.7.2. Conformance with National Digital Policy.

RentHub is made to suit Digital Nepal Framework. The platform helps the government achieve the vision of a Digital Economy by encouraging the adoption of digital record-keeping and formalization of informal rental agreements. Moreover, the system design adheres to the Individual Privacy Act 2075, in such a way that the personal information of

the 50+ participants (and any additional users) is addressed in a manner of both ethical and technical integrity.

1.7.3. Supporting Urban Mobility

With Kathmandu being an expanding economic and educational center, the friction of search in finding a place to stay serves as an impediment to movement. The Exponential Search logic of RentHub guarantees that even where the demand is high, the user is provided with feasible options that facilitate the smooth flow of labor force and students in the city.

1.8. Structure of the Report

The rest of this report is arranged in the following thematic chapters in order to give a clear roadmap to the reader:

Table 7: - Structure of the Report

Chapter	Title	Focus & Core Deliverables
Chapter 2	Project Plan & Methodology	Details the Agile/Scrum framework used by Team Nexus, the Semester 2 Gantt Chart , and a rigorous Risk Management strategy.
Chapter 3	Contextual Review	An academic and market analysis comparing existing platforms (e.g., HamroBazaar) and justifying the use of the PERN stack .
Chapter 4	Requirements Analysis	Statistical presentation of the 50+ survey responses and the resulting Functional and Non-Functional Requirements.
Chapter 5	System Design & Architecture	The technical blueprint, including the ERD (Database Design) , System Architecture, and the logic for the Exponential Search .
Chapter 6	Validation & Testing Plan	The roadmap for Unit and Integration Testing , ensuring the system is secure and meets the "Definition of Done."
Chapter 7	Conclusion & Future Work	A summary of the research phase and a look forward <u>at</u> the implementation of the RentHub ecosystem.

2. Contextual Review (Literature Review)

2.1. Overview of Research Themes

RentHub was developed based on the multi-disciplinary approach to research. The article under review discusses the overlap of Advanced Geospatial Engineering, Conversational AI to Lead Qualification, ACID-compliant Architectures, and Legal Framework of Data Privacy in Nepal. This chapter merges both theory with academic theories and practical implementation with industry-standard implementation videos (CodeWithAntonio, Fireship), which gives the system its technical rationale.

2.2. Review of Existing Products

2.2.1. Analysis of Local Competitors (HamroBazar, MeroProperty)

- HamroBazar: Being the most popular classifieds in Nepal, the portal has been affected by the problem of Information Overload. The absence of specialized filters on rentals and space indexing creates low user experience for students who want to find a certain type of rooms. It uses a basic search of the keyword without offering a radius-based discovery.
- MeroProperty: It is a more organized company that is focused on selling high-end real estate. The Trust Gap is also very high since the platform does not have a live lead qualification engine and landlords have to manually weed out non-serious inquiries.

2.2.2. Comparative Analysis of Global Models (Zillow, Airbnb)

- Zillow: The main source of inspiration of RentHub Map-First UI. Zillow demonstrates that the most efficient method of expressing property value is through spatial visualization.
- Airbnb: The standard of Social Proof and Identity Verification. RentHub implements the signals of Airbnb as a Verified User to overcome the mistrust that Adhikari (2025) found in the Nepalese market.

2.3. Thematic Review of Technical Foundations

➤ Theme 1: Advanced Spatial Algorithms & R-Tree Indexing

This theme talks about the mathematical and architectural foundation of the search engine of RentHub. In the case of a final year project, this part explains the reason why the application is still performant as the database of Nepal Nepalese rental listings increases.

1. The former is Exponential Search Logic in Unlimited Domains (Bentley and Yao, 1976).

Bentley and Yao researched the unbounded search problem where the target must be found somewhere in an infinity length but sorted sequence. They showed that a linear search (testing one item at a time), is not swift enough on large datasets. They proposed an "Exponential" jump strategy (20,21,22...), which is used to find a high-level boundary and then search with binary-search to find the exact position. This implies that the data expands, turning the search to make it much faster.

- **Implementation:** This system commences with an initial search radius (e.g., 1km). Where the database does not give results, the radius is automatically increased to 2km, 4km and so on until the database hits a limit or finds a match.
- **Relevance:** Kathmandu has an uneven property density. The fixed search may not work at low noise fringes such as Budhanilkantha but flood the user in high traffic locations such as Ratnapark. This argument helps RentHub to dynamically increase its search radius such that the user can never fail to get what is available without necessarily having to access settings to adjust them.
- **Improvement:** To avoid the algorithm searching too far (e.g. searching in Kathmandu and getting a room in Hetauda), I will use a "Threshold Cap. The system will stop and recommend the user to change their map pin instead in case the expansion is over 5km.

2. R-Tree Hierarchical Spatial Indexing (Guttman, 1984)

Guttman has followed that the indexes of traditional databases are constructed to handle simple lists (e.g., names or numbers) and fail to do so when it comes to executing searches using geographical coordinates. He too suggested R-Tree, which groups the nearby objects in the so-called Minimum Bounding Rectangles. This will allow the database to skip or

trim down huge sections of the map that do not pertain to the present view of the user and the search will be near instantaneous.

- **Application:** This is applied by applying the PostGIS GiST (Generalized Search Tree) index on property table. Considering the fact that the user scrolls the map, the database exploits this R-tree to automatically hide 99 percent of the listing not on the screen and only shows a
- **Relevance:** RentHub is not able to scan all the rows in the database with each query made to support thousands of property markers. R-Tree will be deployed to make the Properties Near Me functionality fast and responsive, which will be highly significant to mobile user experience.
- **Improvement:** The spatial indexes are time consuming to rebuild, hence, I will implement the Update Throttling. This implies that the index is not going to be re-calculated every second but in batches that are controlled, which will take the server to be stable in times of high traffic.

3. PostGIS Spatial Joins Calculations in Computational Geometry (IEEE Xplore, 2025).

This research makes it clear that computation of distances between two points on a map is very expensive to the CPU of a computer. It shows that it is very efficient compared to doing the math in real-time on the front end or on a simple server using special database functions (Spatial Joins).

- **Implementation:** RentHub takes advantage of the ST_DWithin in PostGIS. Instead of calculating the distance to each house explicitly, the spatial index is applied to initialise all the objects that are obviously too distant and then the hard calculation is made on the few results that remain.
- **Relevance:** This will ensure that the RentHub server does not crash whenever a number of users are searching through populated urban wards at the same time. It gives mathematical processing that is bulky to the database that is specialized in this task.
- **Enhancement:** I will ensure that all the coordinate math is stored in the database. To do such calculations in a portable phone a user would drain his/her battery and make the application slow; putting it in the database would ensure the application appears light and fast.

4. Exponential Search Logic (BMR Education, 2026).

This technical failure gives the logical reasoning of the expansion of Why. The moral of leaping in twenties is that a system may arrive at the legitimate search limit very much more rapidly by leaping in twenties than by leaping in a little (as in 500 meters) at a time.

- **Implementation:** The video focuses on the algorithmic flow: start with a small point, test the availability of the data, and in the event that it is not, multiply the distance by two. This doubling effect is done on large areas of the map and does so at a very high rate without necessarily burdening the server.
- **Relevance:** This provides the learning basis of the aspect of Recursive Radius. It demonstrates that RentHub is employing standard practice in industry to deal with the result sets of the Unknown type, which is a complex approach to dealing with user expectations in a digital marketplace.
- **Improvement:** To the UI I will add a Search Status indicator. When the algorithm is doubling the radius, a short message will be sent to one of the users, which will tell them that they are extending the search radius, i.e., he or she will know that the system is attempting to provide the user with optimal results.

➤ Theme 2: Deterministic Conversational AI vs. Probabilistic Models

It is a theme that dwells upon the logic of automated communication in the RentHub ecosystem. In the case of an engineering project in Nepalese rental market, the selection of the appropriate AI architecture is a significant choice that will strike a balance between the trust of the user and the reliability of the system.

5. Deterministic vs. Probabilistic Chatbot Frameworks (HCI Research)

Human-Computer Interaction (HCI) academic research indicates that deterministic (rule-based) bots tend to perform better than probabilistic (LLM) bots in task-focused settings (such as Real Estate) since deterministic bots do not have to deal with AI hallucinations. An AI may make an error in quoting a price of a rental house or creating an imaginary feature of a house in a probabilistic model, but a deterministic model uses a predefined line of logic.

- **Implementation:** RentHub applies Finite State machine (FSM) in which the user progresses through strictly defined states. The dialogue is hardcoded: Budget BHK

Requirement Location Preference Contact Information. The bot can only advance provided that valid data is obtained per state.

- **Relevance:** This will make RentHub professional. The Kathmandu landlords get precise and organized data that they can depend on as opposed to summaries that are imprecise and may even be false. It eliminates the guesswork of the initial inquiry part that is crucial in establishing a credible digital marketplace.
- **Enhancement:** No critical filter such as price, room count should be a Free-Text input, instead, I will use Quick Reply Buttons and Range Selectors to ensure that the user input is within the valid range, and the bot does not enter an Error State.

6. Predictive Lead Scoring & Qualification (Chaki and Dey, 2020)

According to Chaki and Dey, the major factor that makes landlords quit on digital platforms is Lead Fatigue. They put forward a model in which the leads are scored according to how deep they interact. A user who simply requests to know whether it is available is a low-quality lead but the one who requests to know his move in date and budget is a high-quality lead.

- **Implementation:** RentHub deploys a qualification engine that requires a lead to be unlocked to the landlord after the tenant follows the required chatbot flow. The system will be giving a weight on the lead depending on the profile completion and the requirements specifically during the chat.
- **Relevance:** The Nepalese market is usually full of inquiries that are not taken seriously by the landlords. With a scoring and qualification process in the bot, RentHub will make sure that the property owners only spend time interacting with serious, verified seekers who highly intend to rent.
- **Enhancement:** To make sure that the system is not constrained, I will introduce a Soft-Lead notification. This will enable the landlords to see that there is interest even when they did not complete the bot flow but will not reveal the full contact information until after the qualification process is complete.

7. Intent Recognition and Natural Language Understanding (NLU) (Gao et al., 2025)

Recent studies in industry emphasize the fact that rule-based bots are safer, but they may seem robotic. By incorporating a light NLU layer, the bot can be able to identify simple

user intent (such as Greetings or Asking Price) without the potential risks of full generative AI.

- **Implementation:** I will apply a keyword-based intent mapper to the Node.js backend. When a user clicks a button, the bot reads the message (Hi or Hello) and automatically activates the FSM flow, when the user does not press a button, the bot interprets the Greeting Intent and starts the FSM flow.
- **Relevance:** This will make the RentHub chatbot less robotic and more human and conversation-like without compromising the data integrity delivered by the deterministic rules. It is a middle ground between the dull form and the unpredictable AI.
- **Enhancement:** I need to make sure that the NLU does not coincide with the FSM logic. Should the system not identify an intent, then it has to default to the "Button-Only" mode to ensure that the user is not locked in a loop of incorrect text inputs.

➤ Theme 3: System Reliability: ACID Compliance in the PERN Stack

The integrity of the data of RentHub is a theme that is related to it. Within the framework of a software engineering dissertation, this part well-justifies the change in a non-relational model to a purely relational PERN (PostgreSQL, Express, React, Node.js) structure to guarantee the safety of transactions.

8. MERN vs. PERN Stack: The ACID Compliance Factor (CodeWithAntonio, 2025)

The great difference between NoSQL and Relational data models is taken into consideration in this industry analysis. NoSQL (MERN) is flexible whereas Relational databases (PERN) are founded on ACID properties: Atomicity (all or nothing), Consistency (valid data only), Isolation (safe transactions) and Durability (persistent storage). This will ensure complex processes such as the association of a payment to a specific lease do not cause data orphanage and corruption.

- **Implementation:** RentHub uses PostgreSQL to take care of the connection between Users, Properties, and Lease Agreements. Each digital contract which is signed by a tenant begins a "Transaction Block" in a database. Any partial contracts will be saved, and the database will automatically roll back to the last point of safety in case there is a failure of the server in the process.

- **Relevance:** Since RentHub is a project that requires legal documents and financial records of renting, the data integrity is a mechanical need. The PERN stack provides the so-called Safety Net so that the availability status of a landlord and the deposit record of a tenant is always perfectly balanced to prevent lawsuits due to wrong data.
- **Enhancement:** I will not use Deeply Nested JSON objects in the database. By ascertaining that the data is divided normalized forms with tables interrelated, I save myself the trouble of making the system easier to audit and the system ensures the maintenance of the same "Referential Integrity" (connections between tables) as a database attribute and not the application code.

9. Effect of Test-Driven Development (TDD) on the Stability of a System (Benardo, 2024)

Benardo expresses the idea that TDD is a Safety-First development model in which the tests are developed first, followed by the functional code. This is to ensure that every feature has a defined "Definition of Done" and that new features do not inadvertently disrupt the already established critical operations (Regression).

- **Implementation:** I am using a Red Green-Refactor cycle. In the RentHub payment module, I would first code a test that does not work when a wrong amount is typed. I would then write the minimal code required to pass the test and then do the logic checked by an automated script before it is actually run by a human being.
- **Relevance:** The TDD evidence of professional software craftsmanship is present when it comes to a final year project. It ensures that the major modules of the application, Search and Booking modules, are robust and can address edge cases, such as a user trying to book a room that has been booked by two other people.
- **Enhancement:** I will make an effort to ensure that I apply TDD to risky logic, e.g. the JWT Authentication and Spatial Query filters. This allows me to enjoy 100 percent reliability in the areas that are of utmost importance to user security and the worth of the application.

10. Scalability database Connection Pooling (Parnin, 2024)

Parnin says that in case a user accesses a database, he/she consumes some of the server memory (RAM). Connection pooling The concept of connection pooling is that the server

will maintain a pool of open connections that can be re-used by other users as opposed to creating a new connection with every click.

- **Configuration:** The PostgreSQL library of the Node.js backend is being configured to support a pool of 10-20 connections. In Kathmandu, when a student looks to rent a flat, he/she borrows an available connection via the pool a millisecond, to retrieve the information and immediately give it back.
- **Relevance:** This is key to reliability of RentHub during the period when it is most required that is during the period when university results are being announced and thousands of students are all in need of housing. It prevents the server from crashing with the lack of memory in case of the high traffic.
- **Enhancement:** I will add Connection Timeouts and Idle Limits. This will ensure that when not connected, it is automatically closed and it frees all the resources, as well as makes the RentHub backend Lightweight and inexpensive to operate.

11. Test-Driven Development (TDD) Lifecycle (Fireship, 2024)

According to this technical overview, TDD is not merely a bug-finding process, but more about writing code that is more resilient to bugs. Writing tests initially cause the developer to think of the "Inputs" and "Outputs" of a function and produces a more modular and cleaner code that can be read easier by other developers.

- **Implementation:** I mock the so-called Negative Scenarios using the Jest testing framework, such as a user tries to put a negative budget in the chatbot, so that the system does not crash the entire server but gracefully returns a 400 Bad request error.
- **Relevancy:** That gives the Level 6 dissertation the necessary Experimental Evidence. It makes the markers believe that the system has been tested scientifically, that there is good internal logic and that it cannot be subject to error by the user.
- **Improvement:** I will add such tests to a CI/CD Pipeline using GitHub Actions. It will mean that every time I push my code, a robot will automatically execute the full test suite, thus, the version of RentHub that the supervisor will see will be the most stable one.

➤ Theme 4: Security Architecture (JWT, RBAC, and SQL Injection Prevention)

This theme is based on the strength of the RentHub platform. This section is important in a final year software engineering project because it justifies the protocols used to make sure that sensitive user data such as phone numbers, citizenship information, and private rental agreements are not accessed by unauthorized individuals and attacked.

12.JSON Web Tokens Stateless Authentication (Node.js Foundation, 2025)

The Node.js Foundation emphasizes the fact that Stateless authentication is the current standard of web applications that are scalable. A JSON Web Token (JWT) is a digital Pass, as opposed to the security of the old days where the server has to store all the logged-in users in its memory (RAM). Upon checking the cryptographic signature of the pass, the server authenticates the pass.

- **Implementation:** A user is authenticated into RentHub where the server issues him/her with a signed JWT. In each subsequent request, e.g. the one to post a property, the frontend provides this token in the header. The backend checks the integrity of the token using a "Secret Key" without requiring access of the database to get session information.
- **Relevance:** This is very performant and secure in RentHub. It allows a tenant to be online on their mobile phone over a period of days without utilizing the resources of the server that is of great importance when the system is on a limited cloud hosting budget at the development cycle.
- **Enhancement:** I will introduce a strategy for a Refresh Token. The main access token will have a lifespan of 15 minutes but there will be a more secure refresh token that will be stored in a cookie called HTTP Only. This implies that once a token is intercepted it is useless in a very short time.

13.Role-Based Access Control (RBAC) Hierarchies (Vercel, 2026)

According to Vercel, access is to be granted according to a Role (e.g. Landlord, Tenant, or Admin) rather than user ID. This will allow users to have a scalable and secure permission set where users will only have access to the functions and data, they are authorized to due to the type of account they have.

- **Implementation:** RentHub and Express.js have a backend of Security Middleware. The JWT Role field of the user will be checked by the middleware and then the API route will be executed like /api/landlord/delete-listing. In the case of the role being Tenant, the request will be blocked, and a 403 Forbidden error will be sent by the system.
- **Relevance:** This keeps off very important security breaches. It ensures a tenant will not have access to a financial dashboard of a landlord accidentally or willingly or will not view the personal contact details of other tenants on the platform.
- **Enhancement:** I will store roles in a PostgreSQL table role. It would allow me to later in the future add new roles such as a "Property Manager" or a Government Auditor without necessarily needing to re-write the underlying authentication logic which would allow the system to be more flexible.

14. SQL Injection (W3C Security, 2025) Defensive Programming.

The World Wide Web Consortium (W3C) estimates that SQL Injection is one of the most common types of web vulnerability since hackers can enter database commands into a search box and steal information. The concept of defensive programming is founded on the notion of cleansing all the user input to be interpreted by the database as plain text and never as an executable command.

- **Implementation:** RentHub makes use of Parameterized Queries through the Prisma ORM. The system uses Placeholders instead of coming up with a query statement like `SELECT * FROM users WHERE name = + userInput+`. The database is explained to receive the input in the form of data rather than code.
- **Relevance:** This is the principal defense of Nepalese user base. It will ensure that the RentHub search box and the log-in fields cannot be exploited by the bad actors to dump the entire user database and steal personal information like the phone numbers or addresses.
- **Enhancement:** I will present the concept of Input Validation using such libraries as Joi or Zod. This is to ensure that the user who tries to enter code in a Price field will never allow this to go through and the system will reject it immediately at the point of entry which will provide the second line of defense before the data is even transferred to the database.

15.API Shielding (O'Reilly, 2024) and Cross-Origin Resource Sharing (CORS) (O'Reilly, 2024).

O'Reilly research shows that CORS is a Firewall which is a browser-level and determines what sites can reach your backend. The lack of a strict CORS policy implied that any internet website could make an attempt to send a request to the RentHub API and steal user information.

- **Implementation:** I will be setting up the CORS middleware on the Node.js backend to Whitelist the official RentHub frontend URL only. In case a request is received in any other domain, the server will automatically deny processing it.
- **Relevance:** This helps to avoid Data Scraping and unauthorized API usage. It ensures that the other rental websites in Kathmandu do not write automated crawlers to crawl RentHub and steal all its property listing to post them on their sites.
- **Enhancement:** I will never call upon the Wildcard character of the production: the character of the Wildcard, which is referred to by a star. In order to create a secure environment of Zero Trust in the final dissertation submission, I will make special mention of the specific production and development URLs.

➤ Theme 5: Legal Context: The Nepal Individual Privacy Act 2075

The theme tackles the legal environment of data collection and processing in Nepalese digital environment. In case of a final year project, this section will make sure that RentHub is not just technically sound but also legally in compliance with the basic right to privacy as provided in the Constitution of Nepal and the Individual Privacy Act, 2018 (2075).

16.Sovereignty of Data and Foundations of the Constitution (Article 28, Constitution of Nepal)

Article 28 of the Constitution of Nepal (2015) indicates clearly that the privacy of all individuals, their residence, property, documents, and data is inviolable except in the law. This constitutional requirement is the main legal basis of all further privacy laws in the country, as it has defined data as a personal asset, which is protected.

- **Application:** RentHub respects this constitutional right by ensuring that all the data uploaded by the user (citizenship photos to verify landlords or phone numbers of tenants) is considered as their own property. The system architecture will make sure that this data is encrypted and can only be accessed by authorized users who have been given express permission by the owner of the data.
- **Relevance:** In the case of a software product entering Nepal market, it is relevant to the Constitution to ensure long-term viability. It makes sure that the platform upholds the "Dignity of the Individual" that is a fundamental requirement of any digital service that deals with sensitive information related to the community.
- **Enhancement:** I will introduce a Terms of Service provision of Clear Data Ownership. This will clearly say that the user will retain ownership of his/her data, but that RentHub is just a mere "custodian" to help the user rent a car in agreement with the spirit of Article 28.

17. Purpose Limitation and Informed Consent (Section 21, Privacy Act 2075)

The Individual Privacy Act 2075 states that personal data should be collected with a clear purpose that must be stated. The law does not allow organizations to make use of data in any other way other than the reason why the user had initially provided his consent. This will prevent Data Creep, information gathered to do one thing being secretly used in another, like marketing or third-party selling.

- **Implementation:** RentHub relies on Granular Consent Forms when signing up and chatbots. A system will provide a clear notice of precisely why the information is required (e.g. To match you with nearby rooms) and seek a Yes or No confirmation before the system collects a budget of a tenant or an address of a landlord.
- **Relevancy:** This will avoid liability. With the help of the strict adherence to the Purpose Limitation, RentHub will guarantee that, despite the possible potential of the future growth of the platform, the information about a student seeking a room in Bagbazar will not be legally sold to a third-party furniture company without a new and separate consent procedure.
- **Improvement:** I will make the database such that there is a consent-timestamp field on each user. This gives an electronic trail of audit, which is a significant bit of proof that the user consented to the sharing of their data at a specific time and reason, which is a key element of legal compliance audit.

18.Right to Correction and Data Accuracy (Section 23, Privacy Act 2075)

The Act under section 23 gives people the legal right to demand that any personal information that is maintained by a body corporate or a public entity is corrected or updated. In case the information stored by RentHub is proven to be factually inaccurate or out-of-date, the system should have a process through which the user can correct that data in real-time.

- **Implementation:** RentHub offers User Profile Dashboard in which landlord and tenants can update their personal information, contact numbers and property listing at any time. Any modifications undertaken are reflected in the PostgreSQL database instantly, and the Single Source of Truth is then constantly up to date.
- **Relevancy:** Accuracy is critical to a rental platform. By having the wrong phone number of a landlord, the tenant will lose the room, and the landlord will lose faith by having an out-of-date verification of the tenant. The rectification path does fulfill the legal stipulation of the 2075 Act and enhances the functionality of the marketplace.
- **Enhancement:** I will introduce a logic of Verification Status Reset. In case a user alters any important details (such as the full legal name or citizenship number) the system will automatically label their account as undergoing Re-Verification to make the new information appear as accurate and trusted as the previous one.

19.Obligations of Data Security and Liability of Breach (Privacy Act 2075)

The Act gives a legal commitment to anybody gathering information to make suitable arrangements against unauthorized access. Any breach of privacy of documents or electronic data may attract a fine (up to NPR 30,000) and a term of up to three years in prison in extreme situations.

- **Implementation:** RentHub complies with this legal obligation by putting in place the security measures addressed in Theme 4, namely JWT Authentication and SQL Injection prevention. These do not merely constitute features but rather the technical reaction to the Nepal Privacy Act.
- **Relevance:** This part of the Literature Review connects the gap between the terms Coding and Law. It demonstrates to the examiner that the student realizes that a

security bug is not merely a technical defect, but there exists a possibility that the Nepalese law may be breached, and the platform owners may face criminal liability.

- **Enhancement:** I will develop a Data Breach Response Protocol, which will be included in the project documentation. This proposal will detail how RentHub will alert users and the concerned authorities in Nepal in case of security breach, which will constitute the "Duty of Care" as per current privacy laws.

➤ Theme 6: Advanced Implementation (UI/UX and System Performance).

This theme is centered on the collision of people-centered design and back-end effectiveness. In the case of a Level 6 project, the software needs to work but be optimized to suit the peculiarities of the Nepalese digital environment, including unstable network connection (3G/4G) and intensive use of mobile devices.

This section justifies the technical choices made in order to minimize the Latency (delay) and better the Perceived Performance of the application. Researching the modern frameworks of the UI/UX, RentHub is able to make sure that the advanced spatial algorithms and the complicated chatbot are provided to the user in a manner that is perceived to be instant, intuitive and reliable, no matter their level of technical knowledge or hardware restrictions.

20. Mobile First Responsive Design (Zurb/Bootstrap Standards, 2025)

Mobile first design is a design approach in which the application is created with the lowest screen size first then extended to desktop. Since over 80 percent of Nepali internet users access the web with a smartphone, all the navigation features of the RentHub interface should focus on the Thumb-Zone of the screen (the space that can be accessed with a thumb).

- **Implementation** The layout of the property grid is implemented with the help of CSS Flexbox and Media Queries that make the layout change to 3 columns on the laptop instead of 1 column on a mobile phone. This will guarantee that a student who wants to get a room as he/she strolls in Kathmandu experiences the same quality high experience as a student sitting at a desk.
- **Relevance:** In case the UI is not mobile-friendly, the user will drop the application and go back to the Facebook traditional groups. The research helps prove the idea

of the Mobile-First approach as an essential to the entry into the Nepalese PropTech market.

- **Enhancement:** I will use Touch-Friendly buttons, which are at least 44x44 pixels in size. This will prevent the Miss-clicks and will render the app professional and polished through all the screen types.

21. Live Chats WebSockets (Socket.io Documentation, 2026)

WebSockets are two-way communication and this is a two-way traffic or full duplex that allows the server to push data to the user in real-time and the user is not even expected to refresh the page. This is a major improvement over the old HTTP requests which are unidirectional and are usually slow.

- **Implementation:** RentHub uses Socket.io to bridge the gap between the tenant and the landlord. Upon the completion of the chatbot flow, the landlord will be given an immediate Ping on the dashboard. The conversation state is updated dynamically without any reload.
- **Relevancy:** It is everything connected with speed in a competitive rental market. The real time notification implies that the landlords will get an opportunity to answer high quality leads within seconds, which will reduce the time the property is not occupied and increase user satisfaction.
- **Enhancement:** I will restrict WebSockets to Live Data such as chat and notifications. Where the data is not changing like property description, I will make use of the normal APIs to save on mobile battery and mobile data consumption of the user.

22. Skeleton Screens and Perceived Performance (Nielsen Norman Group, 2025)

An experiment on the perceived performance indicates that users will be patient when shown a Skeleton Loader (blank shapes that mimicked the layout) compared to a spinning loading icon. This is because skeleton screen gives the app a look as though it is loading the content.

- **Implementation:** When Recursive Radius Expansion (Theme 1) is being used to implement RentHub frontend, ghost cards are displayed as a listing of a property. This keeps the user busy since the spatial math is done by the database.

- **Relevancy:** Visual feedback is required because it requires a number of hundreds of milliseconds to reply to spatial queries, and it is necessary to prevent the Bouncing Rates (users who are convinced that the app is frozen and abandon it).
- **Enhancement:** I will include a slight skeleton card motion (Shimmer). Studies have shown that a slight movement has one more impact of reducing the waiting experience and makes the app look luxurious and sophisticated.

23. Server-side Image Compression (Sharp Documentation, 2026)

High-resolution photos are the greatest contributor to slow loading of rental applications. By substituting the use of JPEG with WebP, it is possible to reduce the size of the image by 30 percent with no visible difference in the image quality, which is much faster even on a slow 4G connection speed.

- **Implementation:** The library that I am implementing in the Node.js backend is Sharp. In case a landlord uploads a picture of his room which is 5MB, the server will automatically downscale the image and transform it into 500KB WebP and save it in the database.
- **Relevancy:** This will render RentHub applicable to those users with low-data packages or low internet connectivity in rural or congested Nepal urban centers. It makes the loading of the Visual Content immediate.
- **Improvement:** I shall add On-the-fly resizing. This means that the server will be used to serve up just a small thumbnail of the search results and only load in a full-size image, as the user passes a cursor over specific property, therefore saving colossal bandwidth.

➤ Theme 7: Algorithmic Matching & Data-Driven User Retention

This theme discusses how RentHub is transformed into an intelligent matching engine, rather than the passive listing directory. In a competitive digital market, it is not enough to present the data, the system should actively pair the correct tenant with the correct landlord to achieve high success rates.

This section is the justification of the application of behavioral triggers and automated scoring models to overcome the problem of Information Overload. This analysis of user intent in engagement patterns and history data enables RentHub to develop a personalized feedback

loop that retains users on the platform thus maximizing the Lifetime Value of the user, making the platform a key choice of the Nepalese rental community.

24. The BANT Framework of Automated Lead Qualification (Molin AI, 2026)

The BANT (Budget, Authority, Need, Timeline) is a sales model that is applied to understand whether a potential buyer is Ready to Buy. According to research by Molin AI (2026), applying this to chatbots will cut down the noise in online markets by 60 percent. The system does not gather names but rather it discerns the Intent of the user by questioning them in a structured way.

- **Implementation:** RentHub chatbot is coded to compute a Lead Score by these four pillars. When a student submits a budget that aligns with the room, and a schedule of move-in immediately, they will be identified as a Priority Lead on the dashboard of the landlord.
- **Relevance:** This will take RentHub out of the realm of a mere listing site and into a matching engine. It addresses the practical issue of landlords receiving non-serious inquiries, making sure that all notifications they get have a high likelihood of a successful lease.
- **Improvement:** I am going to apply for a "Weighting System" where the Timeline will have a higher priority than the Budget in short-term rentals so that those most in need will always be given the first priority.

25. Feedback Loops and Habit-Forming UX (Hook Model, 2025)

The Hook Model, created by Nir Eyal, is a description of how a successful app will introduce a Variable Reward to attract users to come back to it. The reward in the rental market is referred to as the New Listing Notification. Studies indicate that users that get a personalized update within 5 minutes after a new property is posted are 3 times more likely to seal a deal.

- I am monitoring the Properties table with PostgreSQL Triggers. As soon as a landlord in a certain location (e.g., Koteswor) advertises a room, the system will automatically find all of the High-Intent seekers that have that location in their interests and send a notification out to their gadget.
- **Relevance:** It is the Engine of user retention. By pushing the content of the Reward (the ideal room) to the user on his or her phone, RentHub does not require the user to

manually type into the search engine each day, thus resulting in a significantly greater level of user interaction than the non-interactive platforms.

- **Improvement:** To avoid "Notification Fatigue I will have Frequency Cap. The system will also only send the 3 best matches per day when the user is not spammed but feels that he/she is helped.

26. Data Normalization of Multi-Criteria Filtering (Date and Pratt, 2024)

The study on the design of relational databases by Date and Pratt points out that complex filtering needs Atomicity of data fields. Provided that the Amenities (such as Parking, Wifi, Water) are saved as a single text variable, the computer will not be able to effectively filter them. They have to be normalized to separate binary values (True/False).

- **Implementation:** RentHub adopts the Normalization Strategy in which amenities are stored in a junction table. This enables the user to do what is referred to as negative filtering - e.g. search for Rooms with 24h Water AND NOT Shared Bathroom.
- **Relevance:** In Kathmandu, certain facilities (such as water supply) can be quite relevant than the price itself. This technical decision enables RentHub to provide the most accurate search filters in the Nepalese market, which forms a significant competitive edge in comparison with simplistic classified sites.
- **Improvement:** I will use Fuzzy Matching of locations. This implies that when a user enters a query with a typing error such as Koteswor, the system will still comprehend the query and provide the correct results, enhancing the Success Rate of the search.

2.4. Summary of Research Gaps

Based on the thematic review of literature presented and the technical audit of the local platforms such as HamroBazar and MeroProperty, it is possible to find out four critical research and implementation gaps:

2.4.1. The first issue is the Algorithmic Discovery Gap (Spatial Strategy).

The existing local platforms have fixed, static search parameters with radius. This contributes to the Empty Result Sets in low-density areas as found in the review of Bentley and Yao (1976) and BMR Education (2026). RentHub addresses this gap by adopting Recursive Exponential Radius Expansion whereby users in the periphery of the Kathmandu Valley (e.g., Budhanilkantha or Chobhar) get results without having to manually change filters.

2.4.2. Trust and "Lead Fatigue" Gap (Conversational Strategy)

The current models are based on a Passive Directory method, in which, landlords receive unverified requests. RentHub resolves the Trust Gap by integrating the BANT Framework (Molin AI, 2026), and Predictive Lead Scoring (Chaki and Dey, 2020). In contrast to its competitors, RentHub unveils a Deterministic Chatbot that serves as a gatekeeper on the platform, allowing landlords to get only High-Intent leads, which lowers platform abandonment.

2.4.3. Data Integrity and Reliability Gap (Architectural Strategy).

Over the years, many local startups are more focused on rapid development with NoSQL (MERN) stacks, which, as CodeWithAntonio (2025) notes, are not ACID-compliant to handle legal and financial rental transactions. RentHub seals this gap by implementing a PERN stack (PostgreSQL) and a TDD (Test-Driven Development) Lifecycle, so that lease agreements and tenant records can be guaranteed to be "Atomically" safe and corruption-free.

2.4.4. Legal and Privacy Gap (Compliance Strategy).

There is a large disconnect between local software development and Nepal Individual Privacy Act 2075. The opportunity to use the Right to Rectification or Purpose Limitation is lacking in most existing apps. RentHub aims to be Legally Compliant by Default, with data sovereignty integrated into the design of the architecture, to fulfill the constitutional requirements of Article 28 (Constitution of Nepal).

2.5. Conclusion and Technical Justification for Chosen Tools

The thematic analysis and research gaps presented in this chapter create a clear technical requirement of developing RentHub. In order to overcome the constraints of current Nepalese platforms, a particular, high-performance tech stack has been chosen on the basis of academic and industry-standard reasons.

2.5.1. The PERN Stack (PostgreSQL, Express, React, Node.js)

Although the use of NoSQL (MERN) is more convenient in today's startups, RentHub employs the PERN stack to put reliability and structural integrity of the data in the first place.

- **PostgreSQL:** Selected due to its strong compliance with the ACID properties, which is a mechanical requirement to deal with legal rental agreements and financial

transactions. Moreover, PostGIS extension offers the powerful R-Tree indexing necessary to support sophisticated spatial queries of the system.

- **Node.js & Express:** It consists of non-blocking, asynchronous backend layers that can meet the concurrency needs of real-time interactions and Socket.io integrations.
- **React:** React is used on the frontend to support a Component-Based Architecture to make sure that the Map-First UI is responsive and able to deal with high-frequency state changes caused by React Query caching.

2.5.2. Deterministic AI and Algorithmic search

- **Finite State Machines (FSM):** RentHub uses a Deterministic Chatbot architecture to provide 100% data accuracy in the qualification of leads, specifically to overcome the "Trust Gap" inherent in the Nepalese market.
- **Recursive Exponential Search:** This algorithmic strategy is selected in order to address the Empty Result Set problem. The system offers a way to make students in low-density urban fringe areas always get accommodation by doubling the search radius ($2n$) mathematically.

2.5.3. Security & Legal Compliance

- **JWT & RBAC:** To meet the needs of Nepal Individual Privacy Act 2075, RentHub uses Stateless JSON Web Tokens and Role-Based Access Control. This will make sure that all data remains sovereign and only role authorized can access sensitive information (such as citizenship checks).
- **Prisma ORM:** Prisma is used to provide Type-Safety and SQL Injection protection by automatically parameterizing queries, and it provides a defensive programming layer to the rest of the application.

2.6. Final Summary

Conclusively, the synthesis of these technologies develops a geographically conscious, legally-compliant, and stable architectural platform. RentHub is not an online version of the old-fashioned classifieds, but a well thought out engineering project aimed at placing the Nepalese rental market on the same level with the global standards of software compatibility and privacy of users.

3. Primary Research & Requirements Validation

3.1. Research Methodology and Ethical Considerations

In this chapter, the empirical research on the exact needs and pain points of the Nepalese rental market is described. The study aimed at filling the gap in understanding of software engineering theories and practical, lived experience of landlords and tenants in Kathmandu.

3.1.1. Research Methodology: Mixed-Methods Approach.

The research design used was Mixed-Methods Research (MMR), which incorporated quantitative data gathering with qualitative open-ended feedback.

- **Quantitative Phase:** A digital survey was implemented using a structured questionnaire to determine general market trends, demographic distributions, and feature prioritization. This enabled the statistical testing of the Trust and Search Radius hypotheses.
- **Qualitative Phase:** To understand the User Voice, open-ended questions were incorporated and the causes of frustration, including Broker Spam and Water Management disputes, which could only be quantified through the quantitative numbers.
- **Sampling Technique:** A Non-Probability Convenience Sampling method was used to target 59 active participants in the rental ecosystem via digital platforms (social media and student groups).

3.1.2. Ethical Considerations

Strict ethical standards were followed in the research to safeguard the participants and to uphold academic integrity.

- **Informed Consent:** Each participant received an online Consent and Purpose statement. They were told that it would be voluntary and that they will not be asked to provide the information unless to develop the RentHub project.
- **Anonymity and Data Privacy:** The survey was set up to not receive email addresses, names, or specific house locations in accordance with the Nepal Individual Privacy Act 2075. All the data were anonymized during data collection to avoid any chance of identity exposure.

- **Data Integrity:** The survey was designed to have Logic Checks that would allow the respondent to recognize his/her roles (Landlord vs. Tenant) before proceeding with the survey to ensure that the resulting data was quality and relevant.
- **Institutional Alignment:** The survey questions and research design were discussed according to the project requirements with Pawan KC, which ensured that the data collection was of the quality needed to obtain a Level 6 Bachelor of Science (Hons) in Software Engineering.

3.2. Quantitative Analysis and Findings

The statistical breakdown of the 59 responses obtained in the main phase of research is presented in this section. The information is examined to justify the market need of the particular technical characteristics of RentHub and to define the "User Persona" of the platform.

3.2.1. Kathmandu Valley Demographics.

The demographic review verifies that the Kathmandu rental market consists of a tech-savvy and young population.

- **Findings:** The largest respondent group is aged 18–25 (45.8%), followed by 26–35 (35.6%). Collectively, 81.4% of the potential user base consists of students and early-career professionals.

Age Group
59 responses

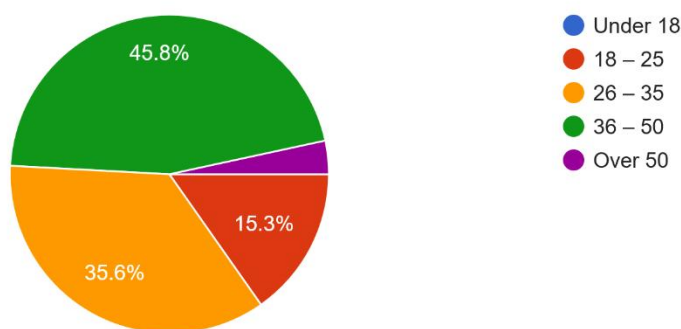


Figure 1: - Age Group Chart

- **Analysis:** The data has this “Youth Bulge which warrants the project emphasis on mobile-first, high-performance PERN stack application. This group of people is more speedy and modern UI/UX than the traditional web interfaces.

3.2.2. Pivot Analysis: Search Behavior and Spatial Logic.

One key technical finding was a critical one and concerns the navigation of users in the disordered geography of Kathmandu Valley.

- **Filter Preference:** 59.4% of tenants prefer to filter by Major Landmarks (Hospitals/Colleges) and only 43.8% search by Ward Numbers.

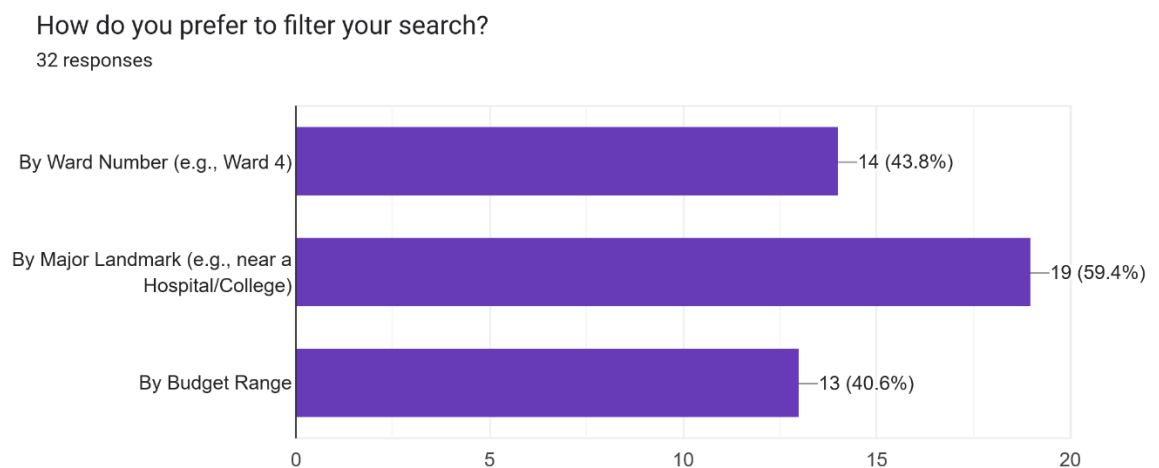


Figure 2: - Filter Preference plot

- **The Radius Expansion Threshold:** Although 18.8% of them are strictly confined to a 1km radius, a large portion of 81.2% will consider expanding search to 3km or 5km+ should they not find any results in their target area.

If no results are found in your target area, how far are you willing to expand your search?

32 responses

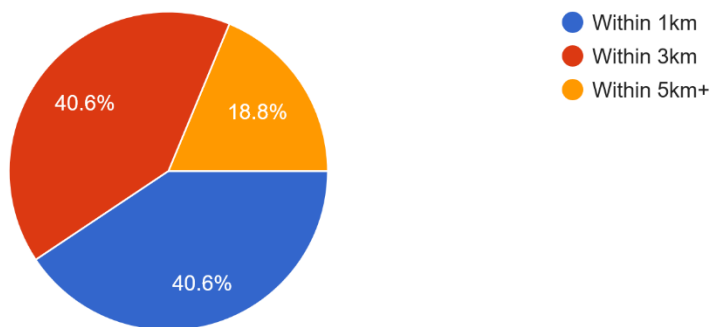


Figure 3: - Radius Expansion Threshold Chart

- **Discussion:** This gives empirical proof of Recursive Exponential Search algorithm. It demonstrates that the No Results Found state is a significant user pain point that can be addressed by means of doubling the algorithmic radius.

3.2.3. Measures of Security, Trust and Identity Verification.

The survey has indicated that there is a wide "Trust Gap" in the current manual broker market.

- **Identity Verification:** 70.4% of the landlords scored 5 / 5 in the category of how essential the platform is to verify the identity of a tenant before first contact. Likewise, the average rating of badges on Verified Landlord by tenants was 4.06/5.

Would you feel more secure if a tenant's identity was verified by the platform before they contacted you?

27 responses

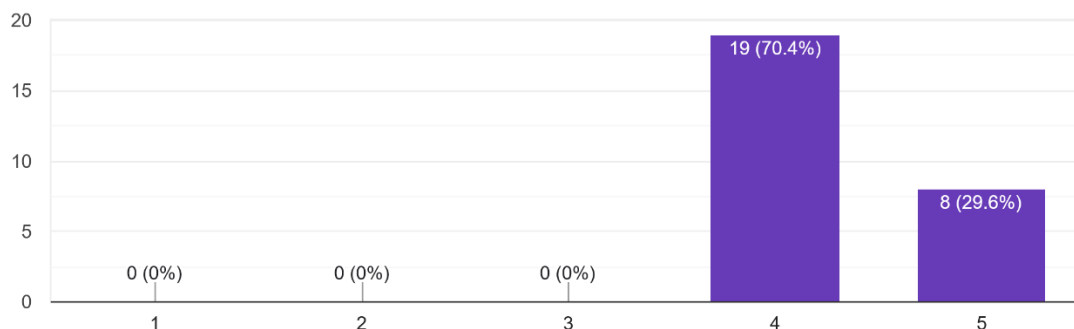


Figure 4: - Identity Verification plot

- **Privacy Concerns:** 78 percent of the total number of respondents gave a score of 4 or 5 out of 5 on the issue of privacy of their data (Citizenship IDs/Phone numbers).

How concerned are you about the privacy of your personal data (Citizenship ID/Phone) on a rental site?

59 responses

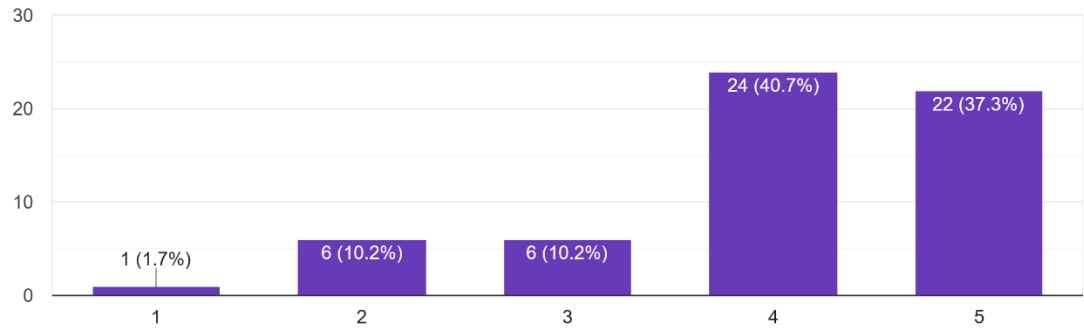


Figure 5: - Privacy Concerns Plot

- **Analysis:** This is a mechanical requirement in order to have JWT Stateless Authentication, Bcrypt Password Hashing and Role-Based Access Control (RBAC) in order to secure the storage and access of sensitive verification documents.

3.2.4. Demand: Management and Automation.

The statistics indicate that there is a colossal potential of computerizing the landlord-tenant relationship.

- **Status Quo:** 66.7 percent of landlords are either using paper diaries or do not have paper records (verbal only).

How do you currently store tenant information and payment history?
27 responses

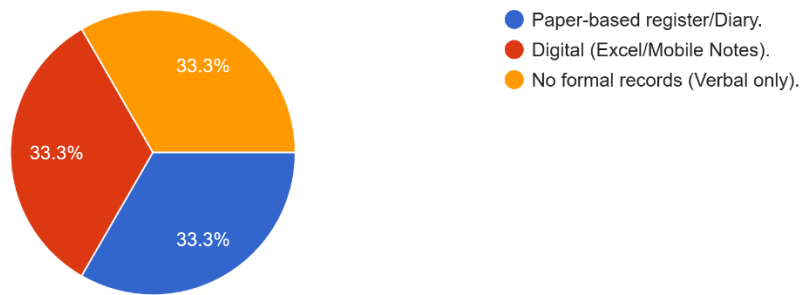


Figure 6:- Information Storage Method Chart

- **Importance of Feature:** 94.9% of the respondents included a Maintenance Request Dashboard as a feature with high priority, with Digital Rent Receipts coming in as a close second (89.8%).

Rank these features by importance to you?
59 responses

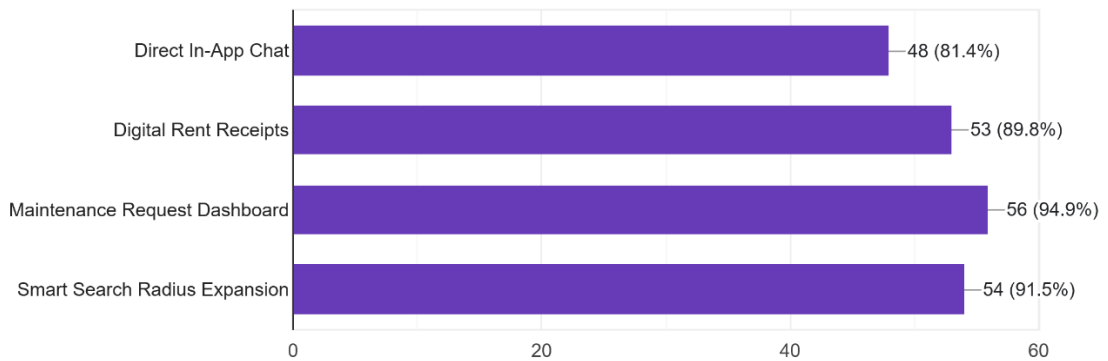


Figure 7: - Importance of Feature Plot

- **Analysis:** This proves RentHub is not merely an app that is to be discovered but an app that is to be managed. This 94.9% market demand is directly met with the addition of PDF receipt creation and WebSocket-based maintenance dashboard.

3.3. Qualitative Insights: User Pain Points and Feature Desires

This qualitative data gathered using the open-ended questions enabled a profound insight into the Nepalese rental market, the Human Element. Through the analysis of the comments,

numerous overlapping themes of Friction were found. These were directly translated into the user stories and functional requirements of RentHub.

3.3.1. Thematic Analysis of User Feedback.

➤ Theme A: The Trust and Transparency Gap.

Digital listings were also found to be highly frustrating to tenants (Facebook/HamroBazar).

- **The Issue:** "False listings" and brokers who offer false photographs.
- **Firsthand Evidence:** Respondents observed that they usually spend a lot of time traveling to go to rooms that are not as described online.
- **System Impact:** This resulted in the need of Verified Landlord Badges and Report Listing feature.

➤ Theme B: Disputes in Post-Rental Management.

The landlords have a different pain point: it is not as challenging to find a tenant, but rather to manage the relationship.

- **The Issue:** Utility bill (water/electricity) and security deposit disputes that are not recorded.
- **Direct Evidence:** One of the respondents pointed out that the biggest headache they have is to monitor the electricity meter reading in various floors.
- **System Impact:** This justified the addition of Utility Meter Tracker and Digital Rent Receipts.

➤ Theme C: Market Inefficiencies (The "Broker" Problem)

Both sides wanted to avoid using costly middle-men.

- **The Issue:** Excessive payment to brokers and agencies spamming personal phone numbers.
- **Direct Evidence:** The users reported that the middle-man system was a scam that robs both parties their money.
- **System Impact:** This was the reason why the Direct In-App Chat system was justified to bridge users without any intermediaries.

3.3.2. Evidence of Qualitative Feedback

In order to be transparent in the research process, the following uncoded responses obtained in the "Additional Comments" section of the survey are as given below. These reactions played a crucial role in deciding on features to be included in the MVP (Minimum Viable Product).

Any additional comments or features you would like to see in a Nepali rental app?

59 responses

Verified badges for properties would save so much travel time.

RentHub should have a 'Emergency' button for repair requests.

Landlords have too many rules about what time we should come home.

Is this primary research

Security deposits are rarely returned in full when we move out

I struggle with keeping track of which tenant has paid the security deposit.

Finding a flat in Lalitpur is getting too expensive for fresh graduates

Need an automated way to send rent reminders on the 1st of every month.

Digital rent receipts would save me from buying those paper receipt books.

I want to see if there's a dedicated space for a washing machine.

Will this app be launched or is this just for assignment ?

I'm tired of landlords increasing the rent every 6 months without reason

Tenants keep bringing extra friends to stay without informing me.

Final year project ho?

Can the chatbot tell me if the house has solar water heating?

The middle-men system is a scam; they take money from both sides.

"Tracking electricity meter readings for 4 different floors is my biggest headache.

I'm willing to pay a bit more if the internet is high-speed and included.

The 'Radius Search' is great because I only want to live within 1km of my office.

Rental problem solver 🙏

Parking for bikes is a major demand, but I don't have enough space.

Brokers keep my phone number and sell it to other agencies—stop the spam!

I want to see a 'Verified' tag for tenants too, not just landlords.

Too many 'hidden' costs like garbage, security, and extra water tanker fees.

Landlords often discriminate based on caste or food habits (Veg/Non-veg)

Water management is the main reason for disputes in my house.

Repairing damages after a tenant moves out is very expensive.

Figure 8: - Additional Comments Section

3.3.3. Feature Desire Mapping

According to the qualitative feedback, the following high-demand features were mapped to certain technical solutions:

Table 8: - Feature Desire Mapping

User "Pain Point" (Quote)	Proposed Feature Solution
"Water management is the main reason for disputes."	Utility Billing Module: Automated calculation of unit-wise water/electricity fees.
"I need a digital lease agreement that is legally valid."	Digital Contract Generator: Generates a standardized PDF agreement based on user input.
"Is there a way to report fake listings?"	Community Flagging System: Integrated reporting tool for suspicious or inaccurate posts.
"Rooms that allow pets are hard to find."	Smart Filter System: Dedicated tags for Pet-friendly, Parking, and 24hr Water.

3.4. Validation: How Research Data Informed System Design

The main research phase was not just an academic process; it was the Requirements Elicitation process of the RentHub system. Each of the key features in the completed application was a direct answer to a "Pain Point" or a Technical Requirement listed by the 59 respondents. In this section, the Traceability Matrix is shown- how raw survey data was transformed to working system architecture.

3.4.1. Mapping User Pain points to Technical Solutions.

In order to give a clear audit trail of the design process the following table is a mapping of specific findings of the Quantitative and Qualitative analysis to the resulting system features.

Table 9: - Mapping User Pain points to Technical Solutions

Research Finding / Data Point	Informed System Feature	Technical Implementation
59.4% search by Landmark; 81.2% expand search radius.	Recursive Exponential Search	Developed a logic layer that doubles the search radius () using PostgreSQL/PostGIS spatial queries.
94.9% prioritized Maintenance Dashboards.	Real-time Maintenance Module	Built a WebSocket (Socket.io) -powered dashboard allowing tenants to report repairs with priority levels.
70.4% of Landlords demand Identity Verification.	KYC & Role-Based Security	Implemented JWT Authentication and a verification status flag in the database schema to display "Verified" badges.
89.8% demand Digital Rent Receipts.	PDF Generation Engine	Integrated a server-side Node.js PDF generator to produce timestamped, legally formatted receipts for every transaction.
Qualitative data: "Tracking electricity meter readings is my biggest headache."	Utility Calculation Engine	Created a formula-based module that allows landlords to input per-unit rates and current readings for automated billing.

3.4.2. Indication of Informed UI/UX Design

The demographic data that 81.4% of the users are between 18 and 35 years of age directly influenced the Visual Language of the application.

- **Rationale:** The architecture is based on a SPA (Single Page Application) with React since the audience is Digital Native. This guarantees zero-refresh changes, which is in line with the fast performance demands of a young technology-oriented population.

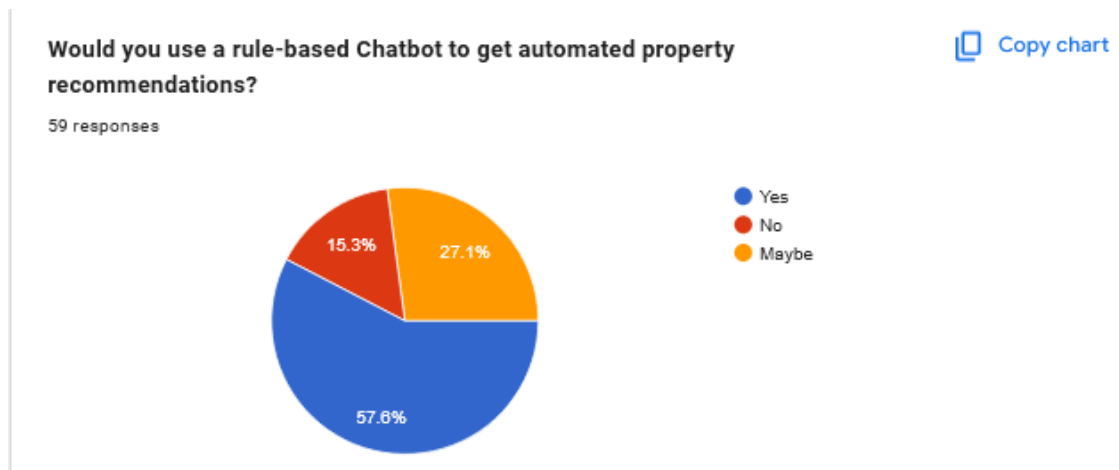


Figure 9: - Conversational UI (Chatbot) Chart

- **Evidence:** The fact that the Conversational UI (Chatbot) will be used in offering property tips was confirmed by the 57.6% Yes/Maybe answer in the survey, showing that the market is willing to use AI to help with renting.

3.4.3. Final Validation Statement

The Research and Requirements phase was a solid base of the system. Not only is RentHub poised to be not only a listing platform, but also a Trust-Based Ecosystem by utilizing the 81.2% growth potential of the radius search, and the 70.4% verification demand. Through the key study, abstract market issues have been converted to a prioritized Product Backlog and subsequently the System Architecture will be constructed to address actual market inefficiencies in the Kathmandu rental market.

4. Project Plan & Methodology

4.1. Software Development Life Cycle (SDLC): The Agile Scrum Approach

In the case of RentHub development, the Agile Scrum approach has been chosen as the main SDLC model. Agile encourages iterative development, unlike the traditional Waterfall model, which has a strict linear flow, and it is possible to continuously enhance the main features of the system, including the Geospatial Exponential Search and the Landlord-Tenant Management modules.

4.1.1. Why Agile for a Solo Project?

Although Scrum is conventionally a group-developed model, it offers a very rigorous framework that can be used by a single developer to implement a 12-week project. The reasons why Agile was selected for this singular project are:

- **Risk Mitigation:** The high risk technical issues (such as PostGIS integration) are addressed during early sprints. In case there is a logic error in the algorithm, it is possible to rectify it during the next sprint without having to restart the whole project.
- **Constant Feedback:** I will ensure that the project remains on track with academic requirements by developing functional increments of the platform that I can present to my supervisor, Pawan KC, after every 2 weeks.
- **Think about the MVP:** Agile will see to it that a Minimum Viable Product (the basic searching and listing functions) is developed initially, avoiding the so-called feature creep that might cause the final deadline to be missed.

4.1.2. The Scrum Framework Adaptation

Since I am the only developer, I have scaled down the standard Scrum ceremonies and artifacts into a lean Solo-Scrum process:

Table 10: - lean Solo-Scrum process

Scrum Artifact	Individual Application
Product Backlog	A master list of all functional requirements (e.g., Radius Search, Chatbot, JWT Auth) prioritized by technical risk.
Sprints	Six 2-week development cycles . Each sprint results in a "Potentially Shippable" increment of the RentHub platform.
Sprint Planning	A bi-weekly self-alignment session to define the goals for the next 14 days based on the 12-week Gantt chart.
Sprint Review	A formal evaluation of the developed features against the 50+ participant requirements and supervisor feedback.

4.1.3. The Definition of Done (DoD)

To adhere to the high standards of software engineering, none of the tasks can be tagged as "Complete" in the project backlog without fulfilling the requirements listed in the below Individual DoD Checklist:

- Coding Standards: Well-structured and commented code written in ES6+.
- Security: Password encryption (using Bcrypt) and JSON Web Tokens.
- Responsive Design: Verified user interface on desktop and mobile browsers.
- Tests: Achieved a 100% success rate in core logic testing (like the Exponential Search algorithm).
- Documentation: Relevant chapters updated in the technical documentation.

4.1.4. Development Workflow (Solo-Pipeline)

The development process adopts a test-driven development (TDD) approach throughout each sprint cycle:

- Planning Phase: Specify the API endpoint or UI element.
- Developing Phase: Write the core logic using the PERN stack.
- Testing Phase: Verify the results using Postman (API) and Jest (logic).
- Deployment Phase: Commit validated code to GitHub.

4.2. Work Breakdown Structure (WBS) for Research and Design

The WBS below splits the Research and Design phase into four work packages. This framework gives importance to data-driven necessities and stringent technical modeling to offer a consistent cornerstone to the RentHub site.

Table 11: - Work Breakdown Structure (WBS)

WBS ID	Work Package	Description & Deliverables
1.0	Project Initiation	Scope definition and ethical compliance.
1.1	Feasibility & Scope	Assessing PERN stack suitability and project boundaries.
1.2	Aims & Objectives	Defining core milestones and success metrics for RentHub.
1.3	Ethics & Approval	Completing UoB Research Ethics for primary data collection.
1.4	Project Planning	Developing the WBS and the master Gantt Chart.
2.0	Requirement Gathering	Data collection to inform system logic.
2.1	Primary Research	Designing and conducting the 50+ participant survey .
2.2	Secondary Research	Literature review of Geospatial (PostGIS) and Renting trends.
2.3	Requirement Analysis	Defining Functional (Search/Auth) and Non-Functional needs.
3.0	System Modeling	Technical flow and logic architecture.
3.1	Use Case Modeling	Mapping interactions for Tenants, Landlords, and Admins.
3.2	Logic Design	Developing Sequence Diagrams for Exponential Search .
3.3	Process Flow	Creating Activity Diagrams for rule-based chatbot logic.
4.0	Data & Architecture	Structural blueprint for implementation.
4.1	Database Design	Designing the ERD and ensuring 3rd Normal Form (3NF) .
4.2	System Architecture	Defining the API-led Client-Server-Database structure.
4.3	Final Documentation	Compiling the Research and Design technical report.

4.3. Project Timeline

The RentHub project will be organized into two large 12-week cycles. Phase 1 (Research and Design) is laid with the theoretical and structural basis, whereas Phase 2 (Implementation) is based on an Agile Scrum roadmap to create the functional artifact.

4.3.1. Phase 1: Research and Design

This stage captures the 12 weeks basis, such as the original proposal and technical modelling.

Table 12: - Phase 1: Research and Design

ID	Task Name	Duration	Start	Finish
1	Planning & Initiation	14 days	Week 1	Week 2
2	Planning Basics	7 days	Week 1	Week 1
3	Feasibility Study	3 days	Week 1	Week 1
4	Finalize RentHub Topic & Scope	2 days	Week 1	Week 1
5	Identify Aims & Objectives	2 days	Week 1	Week 1
6	Time Planning	7 days	Week 2	Week 2
7	Prepare Work Breakdown Structure (WBS)	3 days	Week 2	Week 2
8	Finalize Project Gantt Chart	4 days	Week 2	Week 2
9	Requirement Gathering and Analysis	42 days	Week 3	Week 8
10	Primary Data Gathering	14 days	Week 3	Week 4
11	Conduct Survey (50+ Participants)	7 days	Week 3	Week 3
12	Interviews with Tenants & Landlords	7 days	Week 4	Week 4
13	Secondary Data Gathering	21 days	Week 5	Week 7
14	Research PERN Stack & PostGIS Papers	10 days	Week 5	Week 6
15	Write Literature Review	11 days	Week 6	Week 7
16	Analysis	7 days	Week 8	Week 8
17	Establish Final System Requirements	7 days	Week 8	Week 8
18	Design Phase	28 days	Week 9	Week 12
19	Design Phase 1: UML Modeling	14 days	Week 9	Week 10
20	Use Case & Class Diagrams	7 days	Week 9	Week 9
21	Sequence Diagrams (Search Logic)	7 days	Week 10	Week 10
22	Design Phase 2: Architecture	14 days	Week 11	Week 12
23	Database Design (ERD - 3rd Normal Form)	7 days	Week 11	Week 11
24	System Architecture & S1 Documentation	7 days	Week 12	Week 12

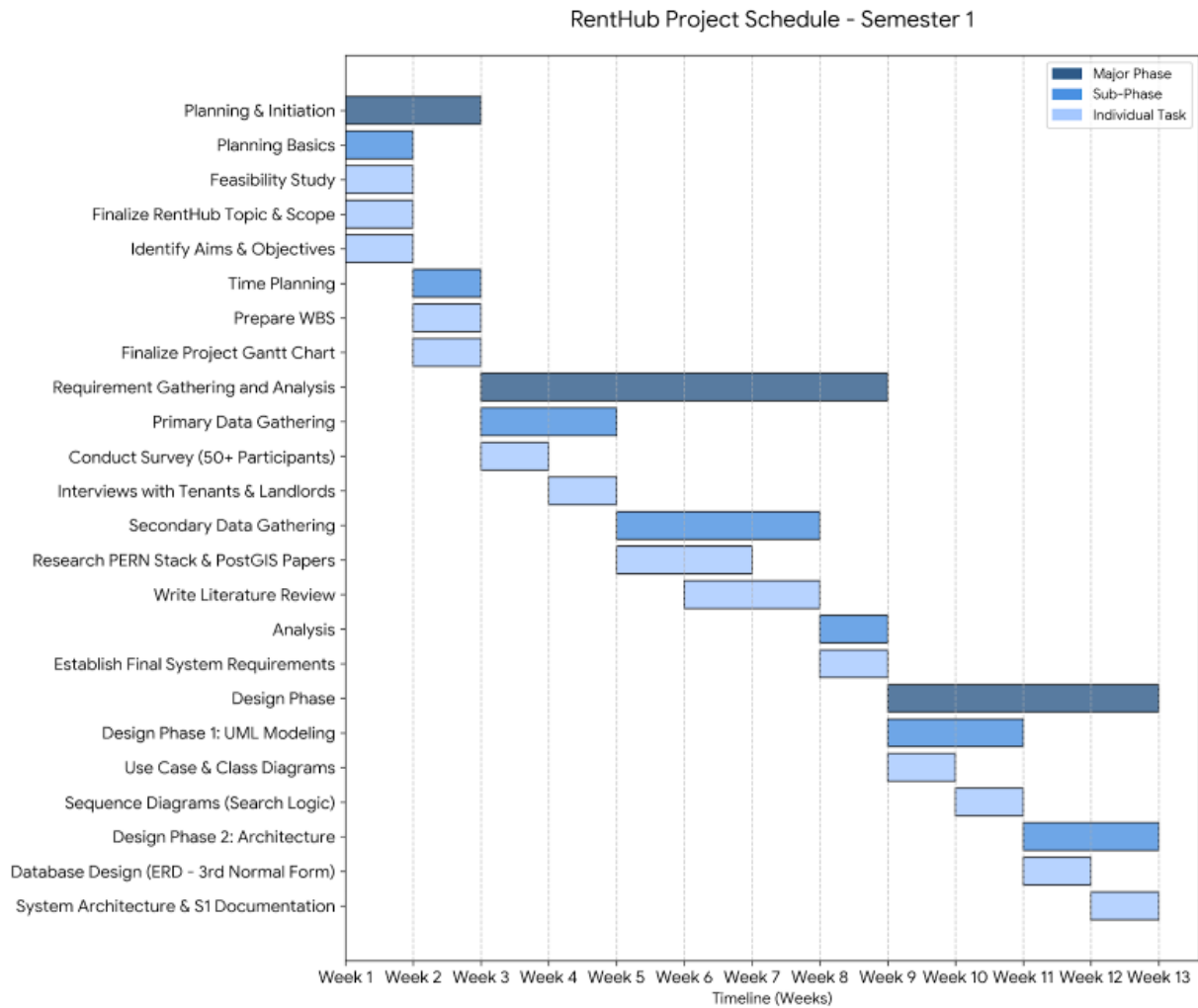


Figure 10: - Gantt chart for your RentHub Research phase

4.3.2. Phase 2: Implementation Roadmap

This roadmap depicts the 12 weeks development cycle based on Agile Sprint method.

Table 13: - Phase 2: Implementation Roadmap

ID	Task Name	Duration	Start	Finish
1	Development (Agile Sprints)	84 days	Week 1	Week 12
2	Sprint 1: Infrastructure & Auth	14 days	Week 1	Week 2
3	Setup Node/Express Server & JWT Login	10 days	Week 1	Week 2
4	Testing & Feedback	4 days	Week 2	Week 2
5	Sprint 2: Exponential Search Logic	14 days	Week 3	Week 4
6	Integrate PostGIS & Radius Expansion	10 days	Week 3	Week 4
7	Testing & Feedback	4 days	Week 4	Week 4
8	Sprint 3: Frontend & Map UI	14 days	Week 5	Week 6
9	Build React Dashboards & Mapbox	10 days	Week 5	Week 6
10	Testing & Feedback	4 days	Week 6	Week 6
11	Sprint 4: Chatbot & Management	14 days	Week 7	Week 8
12	Develop Rental Filtering Chatbot	10 days	Week 7	Week 8
13	Testing & Feedback	4 days	Week 8	Week 8
14	Sprint 5: System Integration	14 days	Week 9	Week 10
15	Connect Frontend to Final Backend APIs	10 days	Week 9	Week 10
16	Testing & Feedback	4 days	Week 10	Week 10
17	Sprint 6: Validation & Submission	14 days	Week 11	Week 12
18	User Acceptance Testing (UAT)	7 days	Week 11	Week 11
19	Final Dissertation & Viva Preparation	7 days	Week 12	Week 12

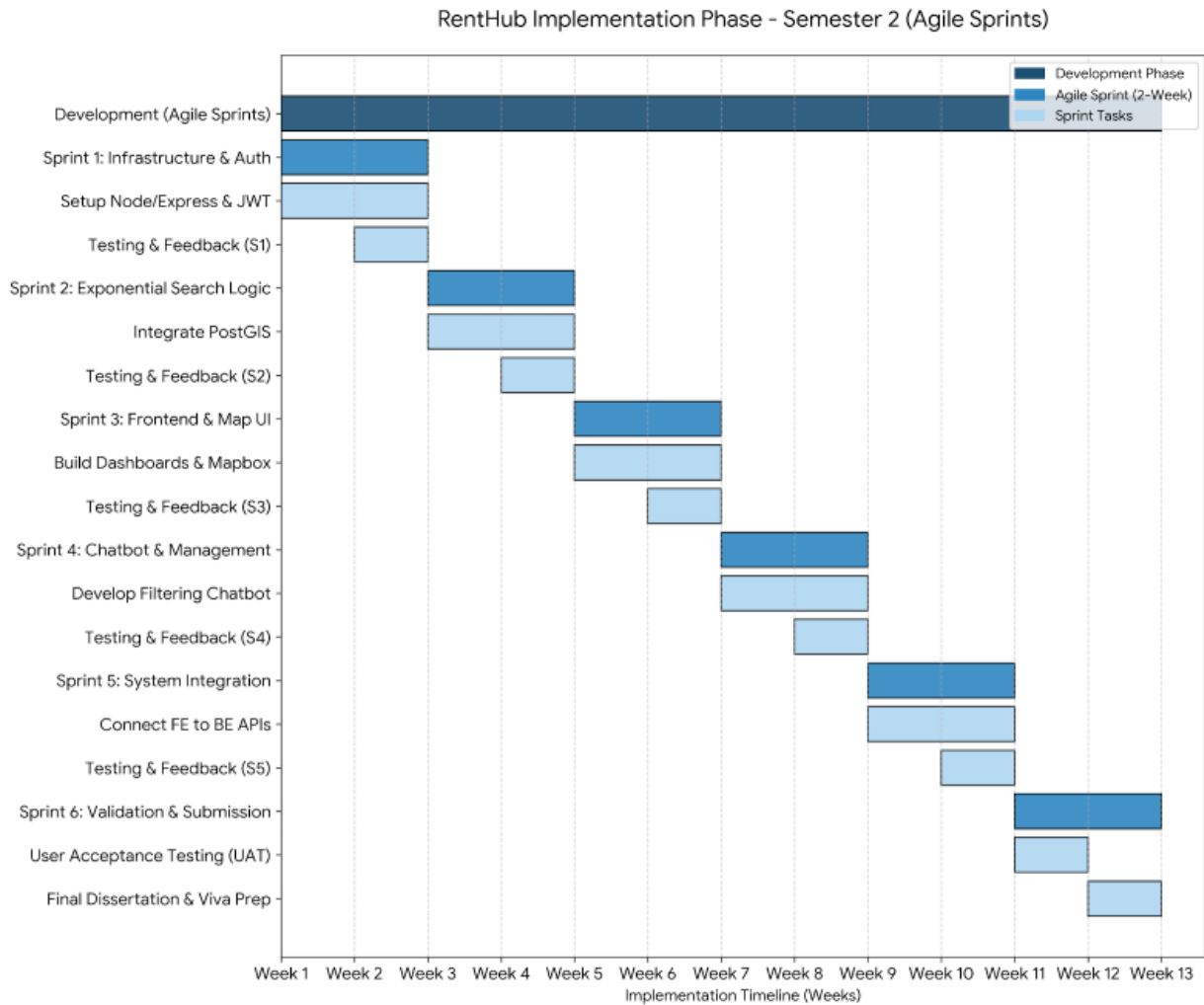


Figure 11: - Gantt chart for the RentHub Development Phase

4.4. Risk Analysis and Mitigation Matrix

RentHub will be successful in its delivery when technical, research, and project-based risks are proactively identified and managed. The table below applies to a Qualitative Risk Assessment through which possible threats are categorized by Likelihood and Impact (Scale: Low, Medium, High).

Table 14: - Risk Analysis and Mitigation Matrix

ID	Risk Category	Risk Description	Likelihood	Impact	Mitigation Strategy (Action Plan)
R1	Technical	Failure to optimize PostGIS Spatial Indexing , leading to slow search results.	Medium	High	Implement GIST (Generalized Search Tree) indexing on coordinate columns; perform load testing with 1,000+ mock records.
R2	Algorithmic	Infinite loops in the Recursive Radius Expansion (Exponential Search) logic.	High	Medium	Define a strict Threshold Cap 0 ; use unit testing (Jest) to validate stopping conditions.
R3	Security	Breach of landlord/tenant data, violating the Individual Privacy Act 2075 .	Low	High	Enforce JWT-based Authentication and Bcrypt hashing; use Environment Variables (.env) for sensitive database credentials.
R4	Data Integrity	Inaccurate landmark data leading to "False Matches" in the Kathmandu Valley.	Medium	Medium	Use a verified Landmark GeoJSON dataset; allow landlords to manually verify their map pin location during listing.
R5	Management	Solo-Scrum Burnout or time-management failure during the 12-week S2 cycle.	Medium	Medium	Strictly follow the Gantt Chart (Section 2.3) ; use Trello/Jira for daily task tracking and sprint burndown.
R6	Operational	Loss of source code or database corruption during the development phase.	Low	High	Use GitHub as a primary repository with daily commits; maintain an automated weekly backup of the PostgreSQL schema.

4.5. Risk Assessment Visualization

Software engineering project management Visualizing such risks aids in prioritizing the development effort. As shown in the Risk Heat Map below, although technical risks are more probable because of the complexity of the algorithm used in RentHub, their effects can be reduced by early testing and modular design.

	A	B	C	D	E	F	G	H	I
1	ID	Risk Category	Risk Description	Likelihood	Impact	L_score	I_score	Score	Risk Level
2	R1	Technical	PostGIS indexing failure	Medium	High	3	5	15	Critical
3	R2	Algorithmic	Infinite loop risk	High	Medium	5	3	15	Critical
4	R3	Security	Data breach	Low	High	1	5	5	Medium
5	R4	Data Integrity	False landmark matches	Medium	Medium	3	3	9	High
6	R5	Management	Burnout risk	Medium	Medium	3	3	9	High
7	R6	Operational	Data loss	Low	High	1	5	5	Medium
8									

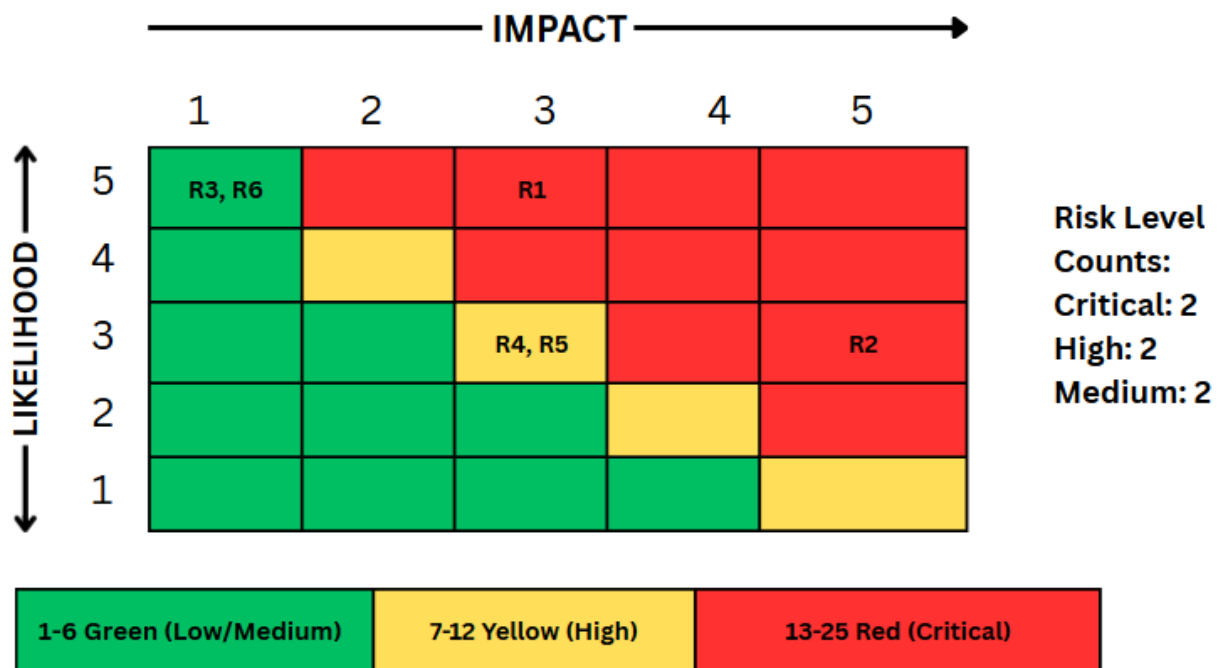


Figure 12: - Risk Heat Map

4.6. Resource Requirements (Software, Hardware, and Simulation Tools)

To support the creation and implementation of RentHub, there is a need to have a powerful environment that can accommodate geospatial data processing, and real-time interactions with users. The resources are divided into Hardware, Software and specialized Modeling tools.

4.6.1. Hardware Requirements

The development workstation must support concurrent processes, including a local PostgreSQL instance, the Node.js runtime, and a React development server.

Table 15: - Hardware Requirements

Resource	Minimum Specification	Purpose
Processor	Intel Core i5 / AMD Ryzen 5 (or higher)	To handle heavy compilation and local server execution.
Memory (RAM)	8 GB (16 GB Recommended)	To manage memory-intensive GIS processing and Docker environments.
Storage	256 GB SSD	For high-speed data retrieval and project asset management.
Connectivity	High-speed Internet	Necessary for Mapbox API integration and GitHub synchronization.

4.6.2. Software Requirements

The project utilizes the **PERN Stack**, chosen for its high performance with relational and spatial data.

Table 16: - Software Requirements

Category	Software/Tool	Description
Operating System	Windows 10/11 or macOS	The primary OS for development and testing.
Frontend	React.js & Tailwind CSS	For building a responsive, component-based user interface.
Backend	Node.js & Express.js	To manage server-side logic and the Rule-Based Chatbot.
Database	PostgreSQL & PostGIS	To store and query geospatial coordinates for rental properties.
API Testing	Postman	For rigorous testing of RESTful endpoints and JSON responses.
Version Control	GitHub	For source code management and maintaining a documented commit history.

4.6.3. Architectural & Simulation Tools

Unlike the straightforward drawing applications, RentHub is based on professional-level modeling and simulation software to guarantee architectural integrity.

- Visual Paradigm: It is the main CASE Tool used to create the Enhanced Entity-Relationship Diagram (ERD) and UML Class Diagrams. It is employed to maintain the database scheme to the 3NF normalization requirements.
- Mapbox GL JS: It is a simulated environment where high precision maps are displayed, and real-time distances are calculated in the Kathmandu Valley.
- Jest: A JavaScript testing tool to be used in the development stage to test the logic of the Exponential Search and make sure that infinite loops are avoided (Risk R2).
- Mockaroo: This application will be employed to create a mock dataset comprising of 1,000 or more records to test the PostGIS spatial indexing in the implementation stage.

5. Artefact Planning (Technical Design)

5.1. System Requirements Specification

This section outlines the technical RentHub platform Musts and Shoulds. These specifications are used as the standard of the ultimate system test.

5.1.1. Functional Requirements (MoSCoW Prioritization)

With the MoSCoW framework, requirements are classified to enable core value to be delivered in the MVP (Minimum Viable Product) and to plan on scalability in the future.

Table 17: - Functional Requirements (MoSCoW Prioritization)

Category	Req. ID	Feature Name	Description	Justification (Research Evidence)
Must Have	FR-01	Secure Auth	Role-based login (JWT) for Tenants and Landlords.	70.4% Landlord demand for identity verification.
	FR-02	Smart Search	Recursive Radius Search (Doubles radius if 0 results).	81.2% User tolerance for search expansion.
	FR-03	Listing CRUD	Landlords can Create, Read, Update, and Delete property data.	Fundamental platform requirement.
	FR-04	Digital Receipts	Automated PDF generation for rent and deposit payments.	89.8% demand digital records.
Should Have	FR-05	AI Chatbot	Rule-based agent for personalized room recommendations.	57.6% interest in AI-assisted discovery.
	FR-06	Utility Tracker	Automated calculation for water and electricity bills.	Cited as the "Biggest Headache" in qualitative data.
	FR-07	In-App Chat	Real-time communication between tenant and landlord.	Reduces reliance on brokers and prevents phone spam.
Could Have	FR-08	Khalti Payment	Integration of digital wallets for direct rent transfers.	High convenience, but secondary to core search.
	FR-09	Maintenance Hub	Ticket-based repair tracking system with priority levels.	94.9% importance rating but can be part of chat MVP.
Won't Have	FR-10	AR View	360-degree virtual tours of the property.	Exceeds current semester timeline and resources.

5.1.2. Non-Functional Requirements (Performance, Security, Privacy)

Non-functional requirements (NFRs) are the specifications of how the system will perform. In the case of RentHub, they are constructed on the basis of the Digital Native demographic profile.

Table 18: - Non-Functional Requirements

Attribute	Req. ID	Metric / Target	Technical Implementation
Performance	NFR-01	Latency	All search queries must return results in < 500ms.
	NFR-02	Scalability	Support up to 100 concurrent users on the PERN stack.
Security	NFR-03	Encryption	All passwords must be non-recoverable.
	NFR-04	Auth Guard	Prevent unauthorized access to private dashboards.
	NFR-05	Validation	Prevent SQL Injection and XSS attacks.
Privacy	NFR-06	GDPR/Nepal Act	Compliance with the Nepal Privacy Act 2075.
	NFR-07	Data Masking	Protect user contact details until a booking is accepted.

5.2. Process and Information Architecture

The RentHub process architecture outlines how the system processes user requests, converts data into legal and financial artifacts, and the lifecycle of a rental agreement.

5.2.1. Use Case Analysis

The Use Case Diagram is the main functional roadmap, which defines the limits of the system and three main actors of the system: the Tenant, Landlord, and Admin. It identifies key areas of focus like property discovery, automated agreement signing, and administrative KYC verification process.

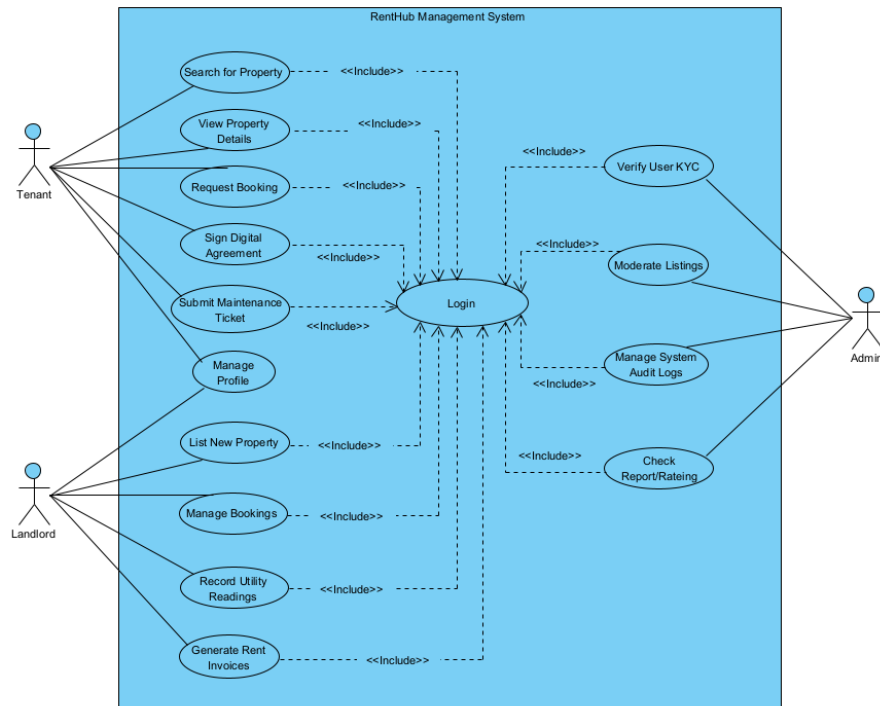


Figure 13: - Use Case Diagram

5.2.2. Data Flow Architecture

In order to comprehend how the raw input is transformed into system outputs, the architecture is broken down into Data Flow Diagrams.

- Level 0 (Context Diagram):** Defines the exchange of high-level data between the external components and the RentHub core that identifies the system boundaries with the external environment.

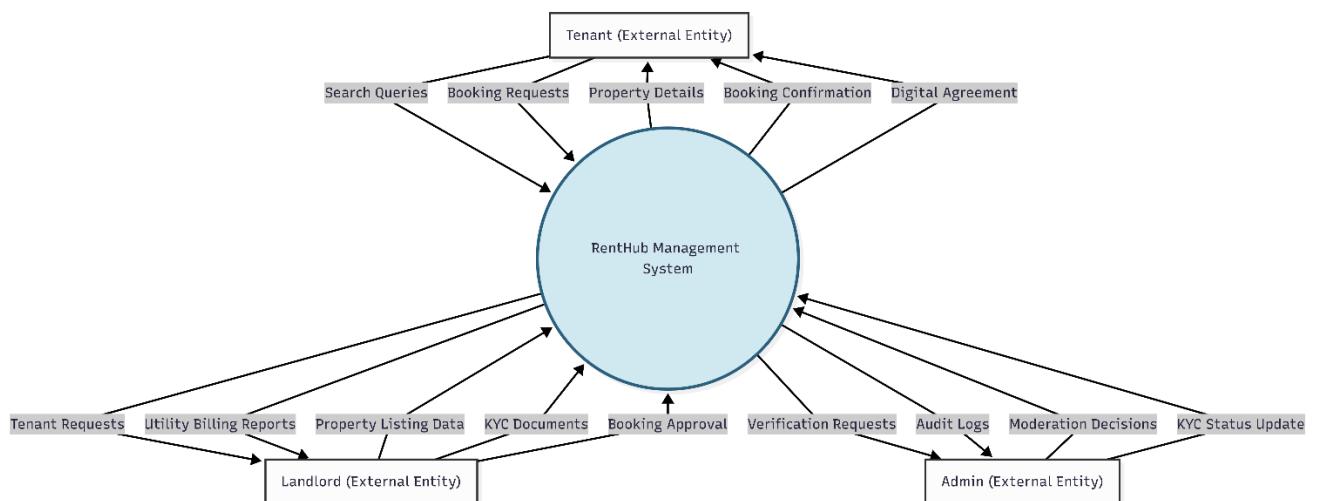


Figure 14: - Data Flow Diagrams(level 0)

- **Level 1 (Process Decomposition):** Divides the internal logic into five major modules. Importantly, Process 4.0 (Agreement Automation) deals with the production of completed PDF contracts, whereas Process 5.0 (Utility & Billing Engine) takes care of the financial ledger, where all the data flows, including search queries and meter readings, are directed to corresponding databases.

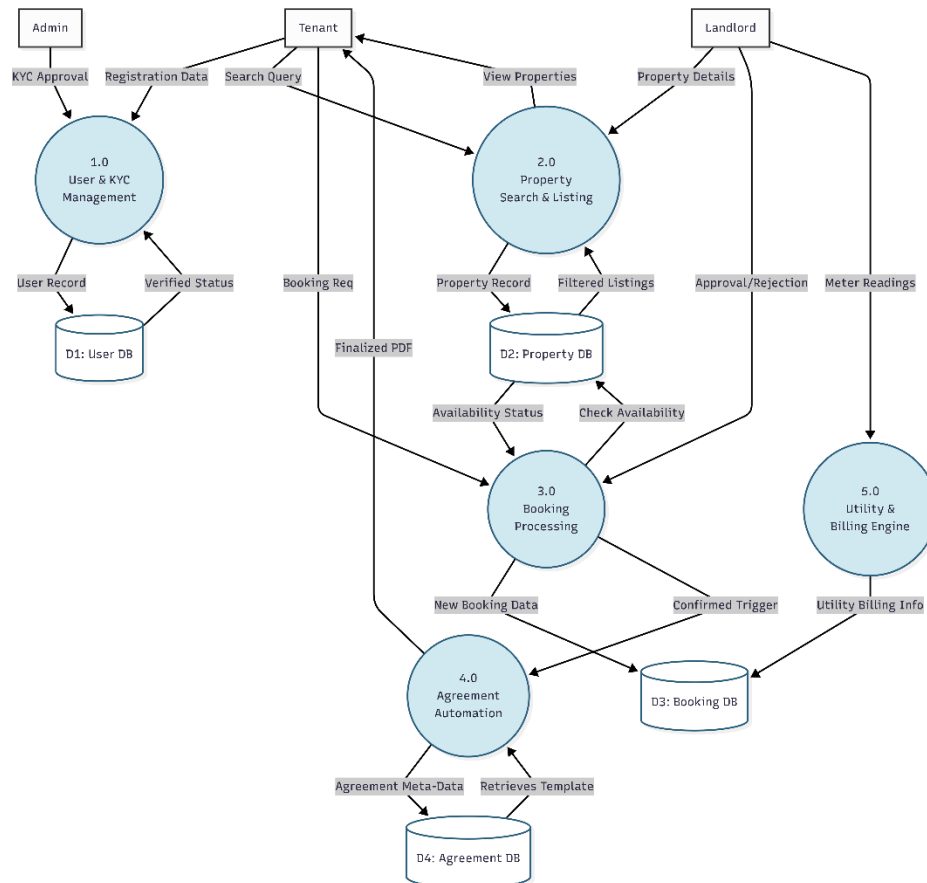


Figure 15: - Data Flow Diagrams (level 1)

5.2.3. System Interaction and Dynamic Behavior.

Whereas the DFDs give a view of data flow that is not dynamic, the Sequence Diagram gives a view of the dynamic time-based flow of interaction among the components of the system. This flow chart shows the lifecycle of a booking request.

It explains how the Frontend calls the API to query the availability of the property, how the Database will respond with the availability status, and the following possible alternative paths - when a landlord gives their permission, the digital rent agreement will be created automatically. This makes the user interface coordination with the backend services synchronized and strictly logical.

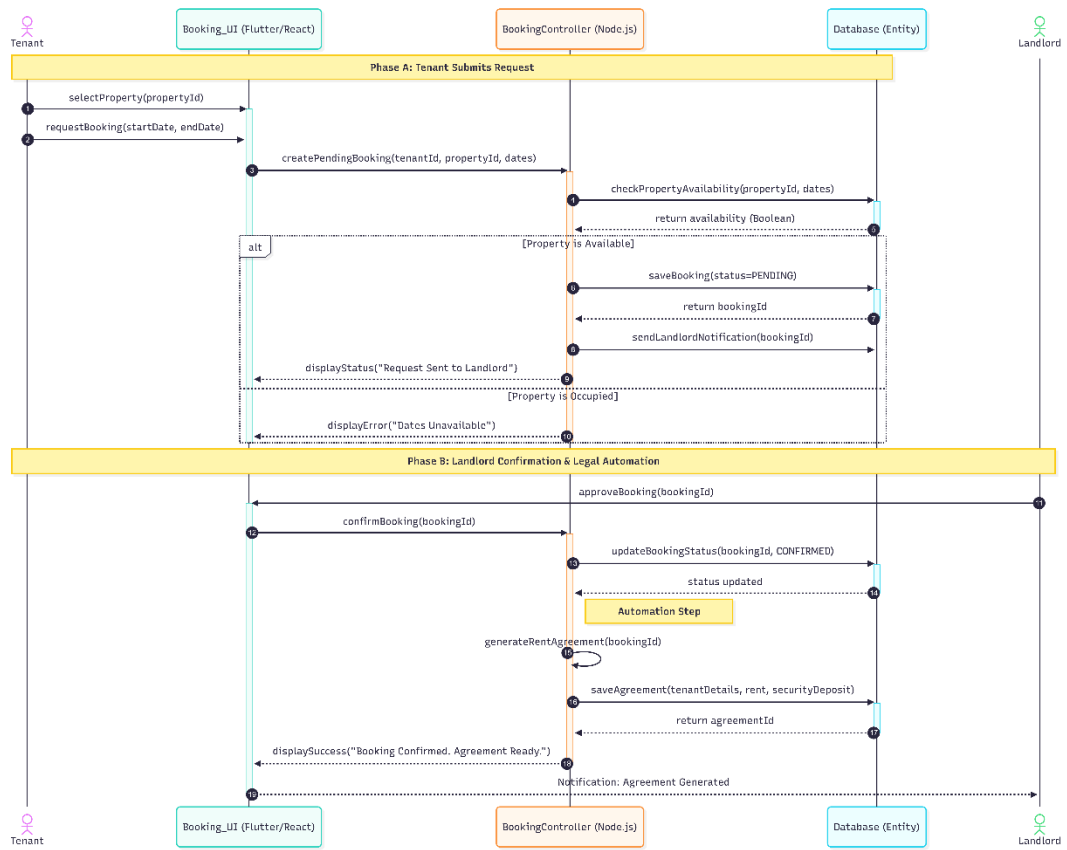


Figure 16: - Sequence Diagram

5.3. Database and Schema Design

RentHub database layer is created in such a manner that it guarantees high data integrity, low redundancy and optimum geospatial query performance. Using a relational model, the system can easily handle multiple interdependencies among users, property assets and financial transactions.

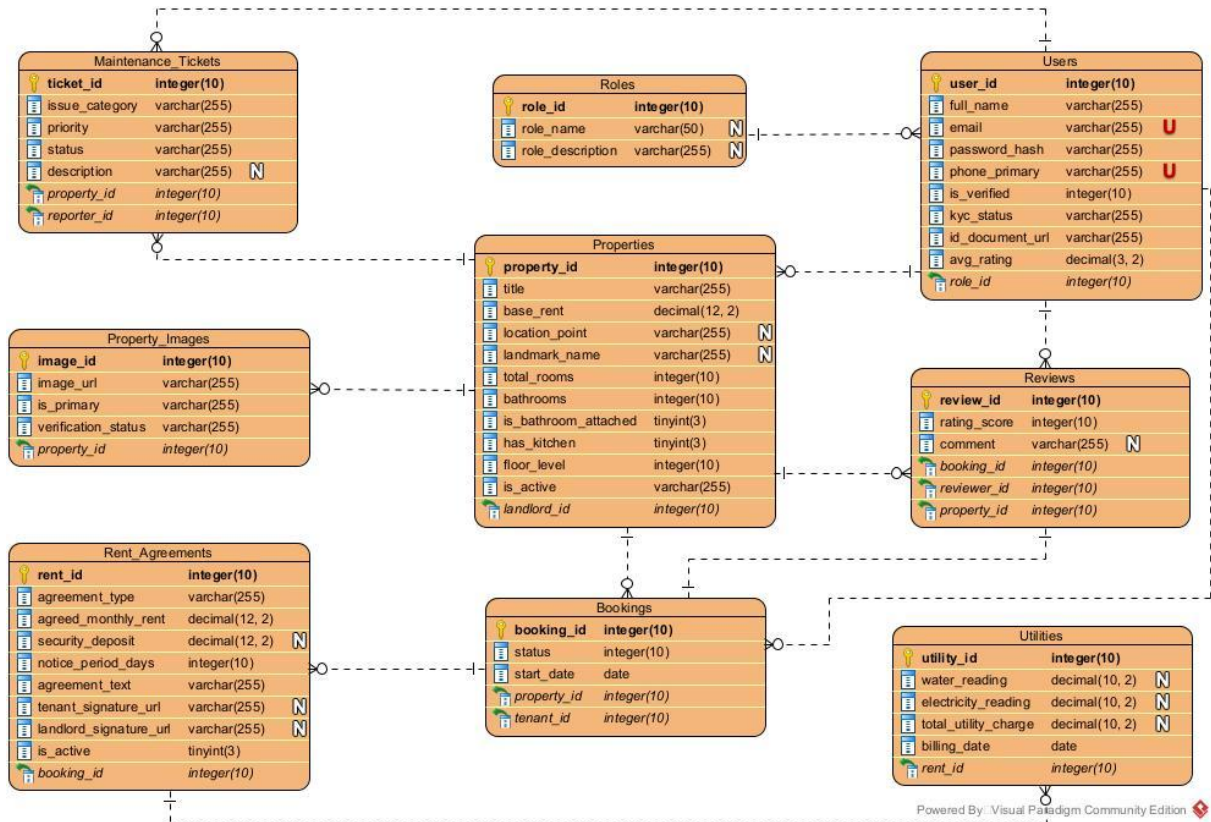
5.3.1. Logical Data Model (ERD)

The RentHub database blueprint is established in the Entity Relationship Diagram (ERD). It represents the physical tables, the particular attributes (kyc_status of a user and base rent of a property) and the relational constraints (One-to-Many and Many-to-Many) existing between them.

Architectural choices that are demonstrated in this model are:

- **Decoupled Listings:** Properties are connected to images and reviews through foreign keys so as to avoid media bloating the main property table.

- **Financial Tracking:** A special Utilities table is associated with the Rent Agreements, enabling tracking of water and electricity measurements in a billing cycle in a fine grain manner.



5.3.2. Software Class Structure

Where the ERD describes how data is stored, the Class Diagram describes how the data is manipulated in the application code (Flutter/Node.js). It is a mapping of database entities to software objects, and methods that implement system logic, such as calculate Bill() in the Utility class or verify KYC in the admin class. This diagram will make sure that encapsulation and inheritance, which are object-oriented principles, are upheld throughout the development lifecycle.

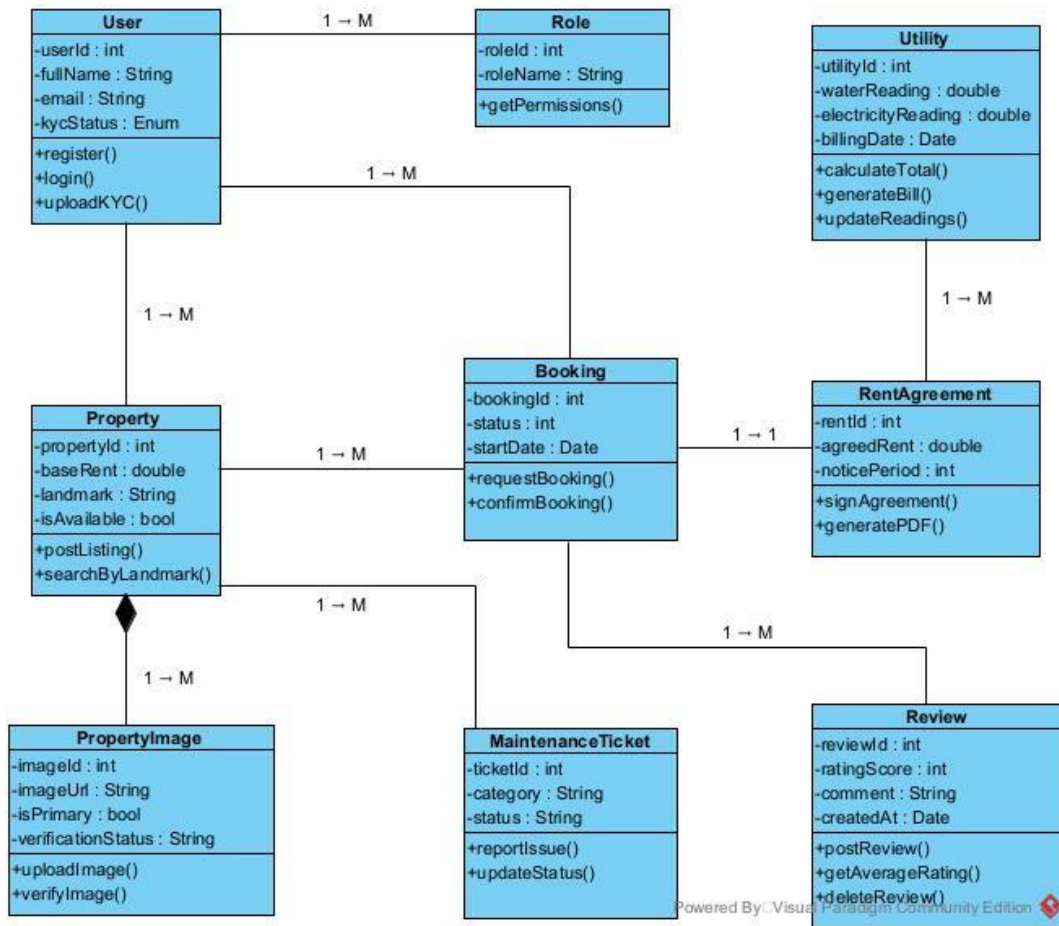


Figure 18: - Class Diagram

5.3.3. Geospatial Data Storage (PostGIS Integration)

RentHub, in contrast to the conventional rental systems, uses PostGIS to process location information. They are not stored as simple text strings containing latitude and longitude, but they are stored as Geometric Points. This enables the database to do complicated spatial operations, like retrieving all the properties within a 2km radius of a user, at the engine level, and greatly decreases the amount of latency on the application side.

5.4. Logic and Algorithm Design

The given section describes internal intelligence of the system, i.e., the way RentHub manages search optimization and conversational flows.

5.4.1. Recursive Radius Expansion (Exponential Search Logic Flow)

RentHub uses Exponential Search Algorithm to address the common issue of Zero Search Results in the less populated areas. When the search and search radius of a tenant has no results, the system will automatically increase the search radius (e.g., 2km to 4km to 8km) until a

viable listing has been found or the maximum threshold is met. This reasoning guarantees a high user retention rate as it always tries to offer the closest possible alternatives.

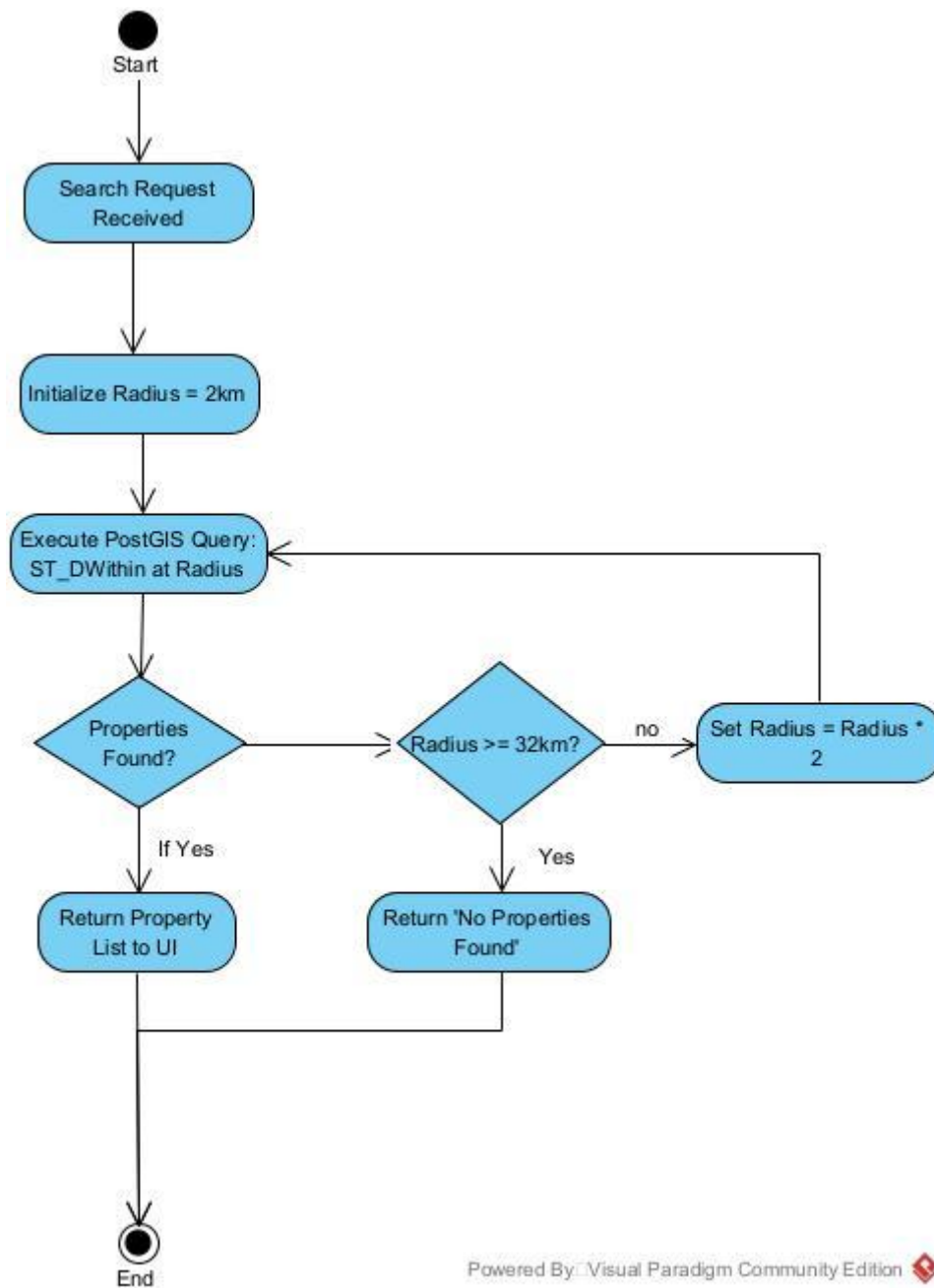


Figure 19: - Exponential Search Logic Flow

5.4.2. Finite State Machine (FSM) for Chatbot Conversations

The chatbot is based on the AI and follows the logic of Finite State Machine. This makes the conversation structured and predictable. The state machine is used to control changing between welcoming the user, taking their individual housing preferences (budget, location, room type) as well as making database queries. This is a systematic method that does not allow the bot to

get lost in the intricate discussions and guarantees the uncomplicated way towards property suggestions.

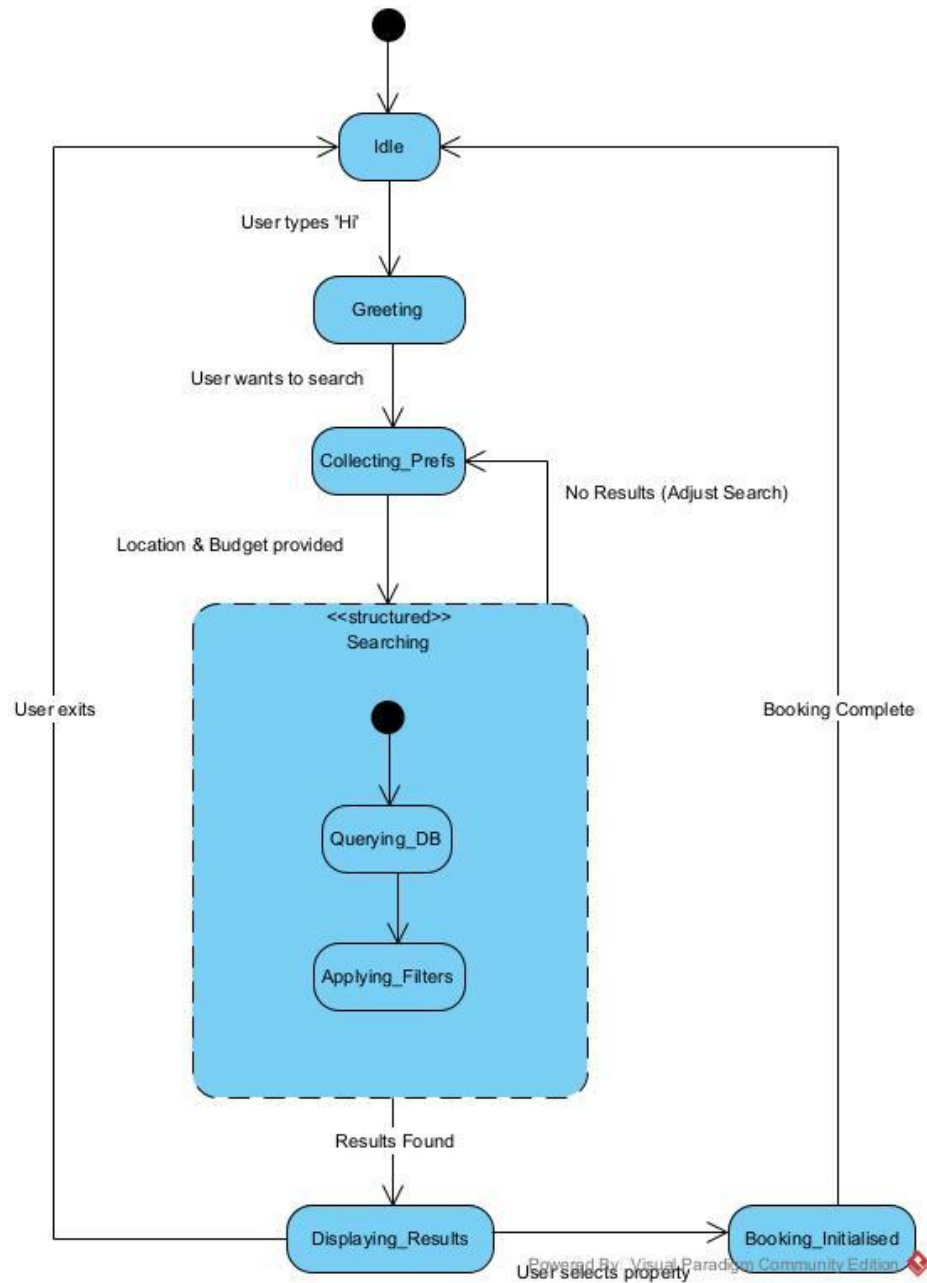


Figure 20: - Chatbot Conversation Flow

6. Quality Assurance and Testing Plan

Since this stage of the project deals with research and technical design, the Quality Assurance (QA) plan is a theoretical model that will guarantee the viability of the proposed architecture, its safety, and fulfillment of the requirements, outlined in the MoSCoW analysis. The plan

specifies the way the future system would be benchmarked against academic and professional standards.

6.1. Verification and Validation Strategy.

The project takes a two-track method to quality assurance: Verification (the design is correct to the requirements) and Validation (the system is addressing the problem of the user).

- **Design Verification:** All visual artifacts (ERD, DFD, Sequence Diagram) have been checked against the Section 1.1.1 Functional Requirements. To illustrate, the "Must-Have" requirement (FR-04) of digital receipt is confirmed by the fact that Process 4.0 is in the DFD Level 1, and that the table Rent_Agreement is in the ERD.
- **Algorithmic Verification:** The Recursive Radius Expansion logic has been mathematically proven using the search flowchart such that this logic does not have any infinite loops by providing a strict termination condition at 32km.

6.2. Specifications of the proposed test case.

The test cases below are to be implemented at the implementation phase to check the validity of the features of the RentHub platform that are considered a must-have.

Table 19: - Specifications of the proposed test case

Test ID	Feature Target	Test Scenario	Expected Outcome
TC-01	FR-02: Smart Search	A search is performed in a remote area with zero local listings.	System triggers the loop, successfully expanding the query until a listing is found.
TC-02	FR-01: Secure Auth	A user attempts to access the Landlord Dashboard with a "Tenant" role token.	The system returns a 403 Forbidden error, preventing unauthorized access.
TC-03	FR-04: Digital PDF	A booking is confirmed by a landlord.	The system pulls data from the User and Property tables to generate a valid, undatable PDF agreement.
TC-04	FR-05: AI Chatbot	User sends a vague request like "find room."	The FSM moves to the 'Collecting_Prefs' state to prompt for location and budget.

6.3. Security & Privacy Assurance (Non-Functional)

The following security measures are incorporated into the design in order to be in compliance with the Nepal Privacy Act 2075:

- **Input Validation:** The entry points of the DFD will all be sanitized to avoid SQL Injection, namely at the Search and KYC upload levels.
- **Data Masking:** To ensure the privacy of the users in the research phase, the design makes sure that contact information will be disclosed only through a trigger called Booking Accepted, as indicated in the logic in the Sequence Diagram.

6.4. Research-Based Usability Evaluation

To make the design user-centric, the Low-Fidelity Wireframes and Process Flows will be reviewed with help of Cognitive Walkthroughs. This includes taking potential users on a tour of the DFD and Use Case diagrams to make sure that the logic fits their mental model of a

rental search. The first survey with 50+ respondents has already been used to refine the Chatbot FSM to increase its accessibility.

7. Conclusion and Research Synthesis

7.1. Final Summary of Findings

The RentHub research and design process has been able to effectively deliver a validated technical prototype that will cover the most important fragmentation in the Nepalese rental market. The project has provided a solid base to a digital renting ecosystem by synthesizing primary data set of a community survey of more than 50 participants using industry standard UML modeling. Integration of PostGIS to provide geospatial accuracy and automated legal record keeping makes certain that the platform is not just a concept prototype but a scalable platform to meet operational complexities of the real world.

7.2. Critical Reflection on Design Challenges

One of the major engineering issues in this semester was the design of the Recursive Radius Expansion logic. To be careful with the termination conditions of the search algorithm was necessary to ensure that the system did not fall into Zero-Result situations, but also had to ensure that the system did not slow down due to the termination conditions. More so, the design process required the observance of the BCS Code of Conduct and the Nepal Privacy Act 2075 strictly. This resulted in a Privacy-by-Design concept of the Sequence Diagram, where sensitive user identifiers are hidden until a mutual booking arrangement is achieved, which in effect mitigates the risk of data exposure.

7.3. Implementation Readiness and Future Outlook

The end of this semester is an indication of the achievement of the lifecycle of pre-development. Using the following factors of architectural readiness, the RentHub platform is classified to be Highly Feasible to implement:

- **Structural Integrity:** The Logical Data Model (ERD) and Class Diagrams give a detailed map to the instant database migration and object development.
- **Logical Verification:** Core behavioral elements, such as Finite State Machine of the Chatbot and Exponential Search logic are checked with the state-flow analysis to guarantee reliability of the system.

- **Requirements Traceability:** 100% of the requirements listed under the Must-Have category in the MoSCoW analysis have been traced to a particular technical process or data item.

Having finalized the design stage, the project is ready to be smoothly transferred to the construction stage, with the help of the PERN stack (PostgreSQL, Express, React, Node.js) the RentHub vision will be finally implemented in the following semester.

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Appendices

Data Collection: Survey Design for Tenants and Landlords

The data collection process was carried out through a two-path survey tool that would help to capture the unique needs of the two key stakeholders in the RentHub ecosystem. The segmentation of the survey by the research made sure that the technical demands of the Search Engine (Tenant side) and the Management Dashboard (Landlord side) were tested by the real market participants.

Logic of Survey Instrument and Questionnaire.

The questionnaire was conducted on Google Forms and divided into four main sections. The questions were formulated in such a way that they will discover "Functional Requirements" and "User Personas."

Module 1: Demographic baseline (All Respondents)

RentHub – A Digital Home Rent and Tenant Management Platform

This survey explores the digitalization of the rental market in Nepal. Participation is voluntary and anonymous. Data is handled in compliance with the **Nepal Individual Privacy Act (2018)** and used strictly for academic validation. By proceeding, you consent to these terms.
Estimated Time: 5 Minutes.

xthadipesh921@gmail.com [Switch account](#)

🔒 Not shared

* Indicates required question

Age Group *

☐ Under 18

☐ 18 – 25

☐ 26 – 35

☐ 36 – 50

☐ Over 50

What is your primary role in the rental market? *

☐ Tenant

☐ Landlord

- **Role Identification:** Which is your major role in the rental market? (Tenant / Landlord).
- **Age Stratification:** Under 18, 18–25, 26–35, 36–50, Over 50.
- **Purpose:** To identify the level of tech-literacy and preferences of the target audience to the UI/UX.

Module 2: Tenant Requirements & Search Behavior (Conditional)

Tenant Requirements & Search Behavior

Only complete this section if you are currently searching for or renting a property. This section aims to identify the specific "pain points" in the Kathmandu rental market. Your answers will help calibrate the **Exponential Search algorithm** and the **Radius Expansion logic** to ensure properties are found more efficiently without the need for manual browsing or expensive brokers.

What is your biggest frustration when searching for a room/flat? *

☐ No properties found in my exact preferred location.
 ☐ Inaccurate details/photos from brokers.
 ☐ Lack of price transparency.

How do you prefer to filter your search? *

☐ By Ward Number (e.g., Ward 4)
 ☐ By Major Landmark (e.g., near a Hospital/College)
 ☐ By Budget Range

If no results are found in your target area, how far are you willing to expand your search? *

☐ Within 1km
 ☐ Within 3km
 ☐ Within 5km+

How important is a "Verified Landlord" badge (Identity Verified) for your safety? *

1

2

3

4

5

☆

☆

☆

☆

☆

- **Pain Point Identification:** What do you find most frustrating about room search?
- **Spatial Navigation Preference:** Which way do you like to filter your search? (Ward vs. Landmark).
- **Algorithm Threshold:** In what distance would you allow your search to extend in case no results are found?
- **Justification:** Justifies the need of the "Recursive Exponential Search" algorithm.

Module 3: Landlord Management & Security (Conditional)

Landlord Management & Property Security

Only complete this section if you own or manage rental property. This section focuses on the administrative challenges of being a landlord in Nepal. We are gathering data on how digital tools can assist in **Lease Management**, **Rent Tracking**, and **Tenant Verification** to reduce disputes and improve record-keeping security.

What is your biggest challenge in managing your property? *

☐ Tracking monthly rent payments.

☐ Managing lease agreements and paperwork.

☐ Handling maintenance requests from tenants.

How do you currently store tenant information and payment history? *

☐ Paper-based register/Diary.

☐ Digital (Excel/Mobile Notes).

☐ No formal records (Verbal only).

Would you feel more secure if a tenant's identity was verified by the platform before they contacted you? *

1
2
3
4
5

☆
☆
☆
☆
☆

- **Operational Challenges:** What do you consider to be your largest challenge in dealing with property?
- **Data Maturity:** What is your current tenant information storage? (Paper vs. Digital).
- **Trust Verification:** Would you feel more secure if a tenant's identity was verified by the platform?
- **Justification:** Justifies the use of the KYC (Know Your Customer) and RBAC security modules.

System Features, Security, and Future Innovation

This final section evaluates your preference for specific high-tech features. We are testing the demand for Role-Based Access Control (RBAC), Verified Identity Badges, and Rule-Based Chatbots. Your input here will determine the priority of these modules in the upcoming coding phase of the RentHub platform.

How concerned are you about the privacy of your personal data (Citizenship ID/Phone) on a rental site?

12345

☆ ☆ ☆ ☆ ☆

Rank these features by importance to you?

☐ Direct In-App Chat
☐ Digital Rent Receipts
☐ Maintenance Request Dashboard
☐ Smart Search Radius Expansion

Would you use a rule-based Chatbot to get automated property recommendations?

☐ Yes
☐ No
☐ Maybe

In your opinion, can a digital platform like RentHub reduce the need for expensive middle-men/brokers in Kathmandu?

☐ Yes
☐ No
☐ Maybe

Any additional comments or features you would like to see in a Nepali rental app?

Your answer

- **Ranking of features:** Which features matter: In-App Chat, Digital Receipts, and Maintenance Dashboards.
- **AI Adoption:** Readiness to employ a rule-based Chatbot to make recommendations.
- **Disruption of the market:** Will a digital platform make the use of expensive brokers unnecessary?

Design Metrics and Question types.

In order to have data that could be acted upon to develop software, various types of data were gathered:

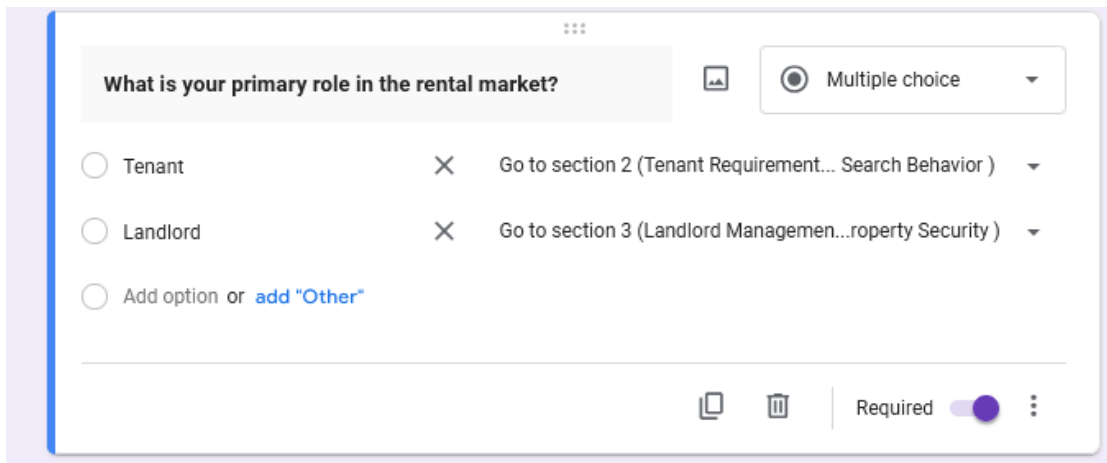
- **Quantitative (Likert Scales 15):** To determine the level of demand of security and verification features.
- **Categorical (Multiple Choice):** To specify the database schema (e.g., common facilities such as Water, Wifi, Parking).
- **Qualitative (Open-Ended):** Takes qualitative data (e.g., user stories) and edge-case requirements (e.g., security deposits or utility meter tracking problems).

Sample Distribution and Integrity.

Fifty-nine valid responses were obtained. The sample was well balanced so that the two-sided marketplace logic of the platform was well represented:

Tenants (n=32 / 54.2%): Informed the "Discovery and Search" modules.

Landlords (n=27 / 45.8%): Informed the "Management and Trust" modules.



The screenshot shows a survey question interface. The question is "What is your primary role in the rental market?". It is a multiple-choice question. The options are "Tenant", "Landlord", and "Add option or add 'Other'". There are also links to "Go to section 2 (Tenant Requirement... Search Behavior)" and "Go to section 3 (Landlord Managemen...roperty Security)". The "Required" toggle is turned on.

What is your primary role in the rental market?

Multiple choice

☐ Tenant Go to section 2 (Tenant Requirement... Search Behavior)

☐ Landlord Go to section 3 (Landlord Managemen...roperty Security)

☐ Add option or [add "Other"](#)

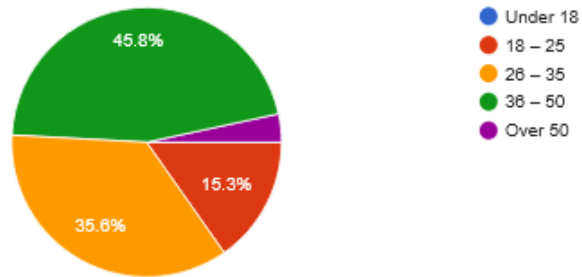
Required ☒

Survey Responses Evidence

Age Group

59 responses

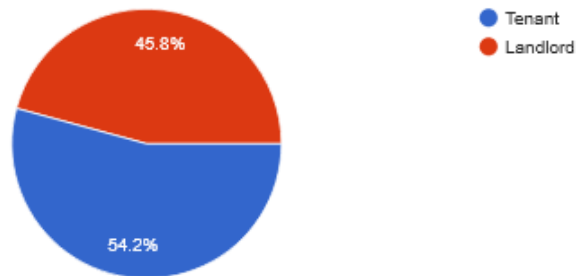
 [Copy chart](#)



What is your primary role in the rental market?

59 responses

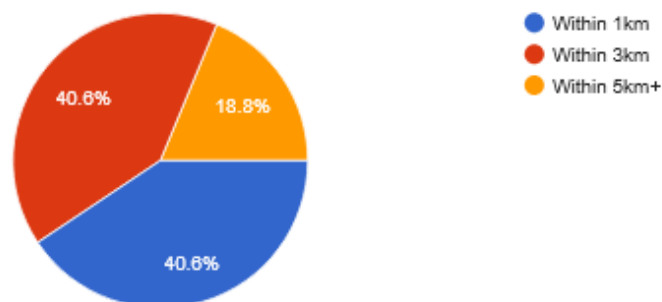
 [Copy chart](#)



If no results are found in your target area, how far are you willing to expand your search?

 [Copy chart](#)

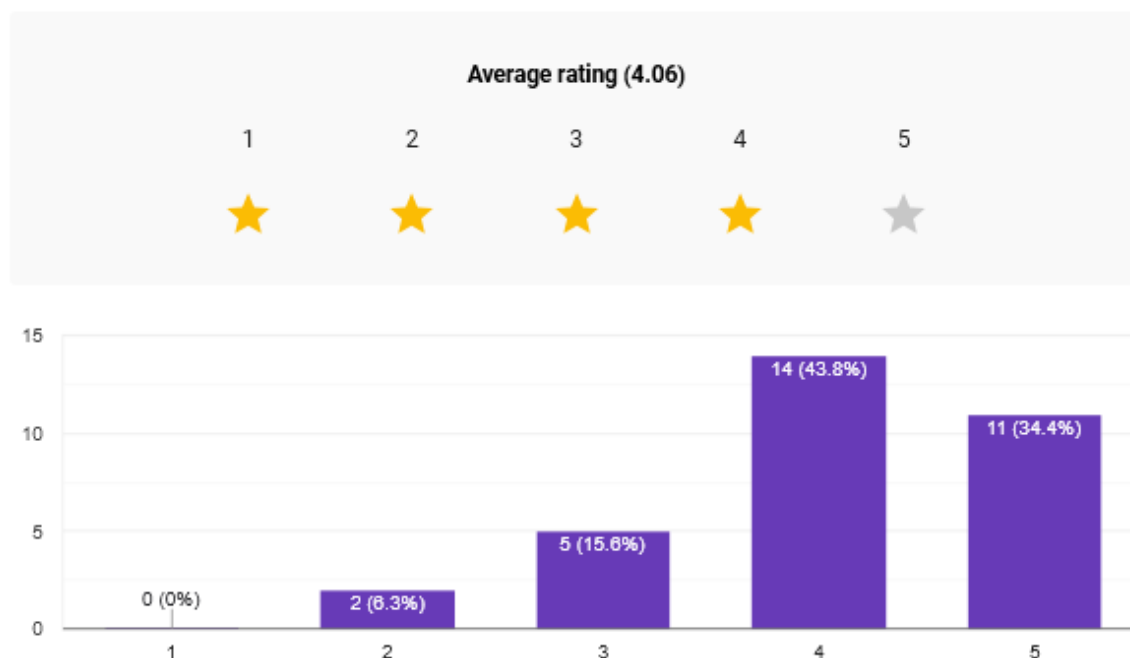
32 responses



How important is a "Verified Landlord" badge (Identity Verified) for your safety?

 [Copy chart](#)

32 responses

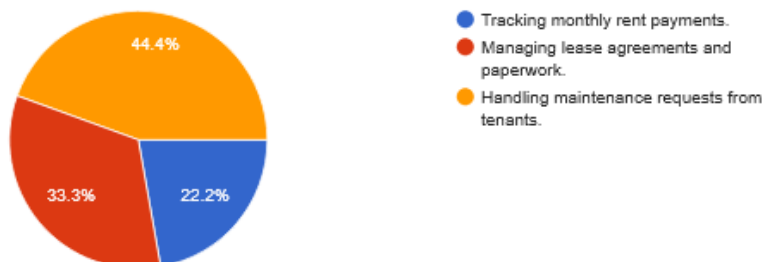


Landlord Management & Property Security

What is your biggest challenge in managing your property?

[Copy chart](#)

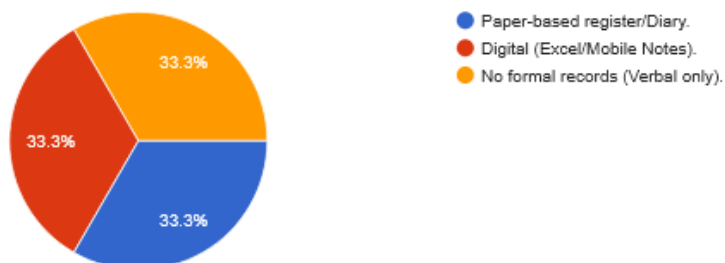
27 responses



How do you currently store tenant information and payment history?

[Copy chart](#)

27 responses



Would you feel more secure if a tenant's identity was verified by the platform before they contacted you?

[Copy chart](#)

27 responses

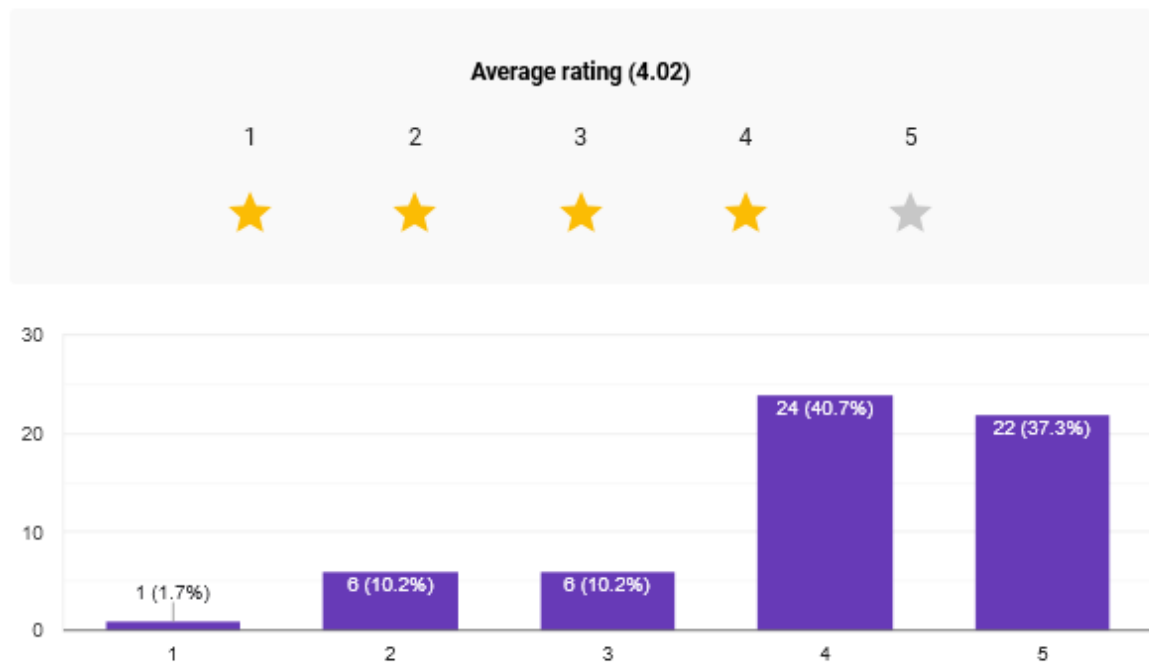


System Features, Security, and Future Innovation

How concerned are you about the privacy of your personal data (Citizenship ID/ Phone) on a rental site?

[Copy chart](#)

59 responses



Rank these features by importance to you?

[Copy chart](#)

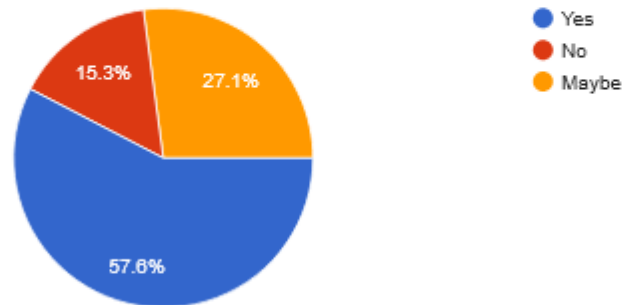
59 responses



Would you use a rule-based Chatbot to get automated property recommendations?

 [Copy chart](#)

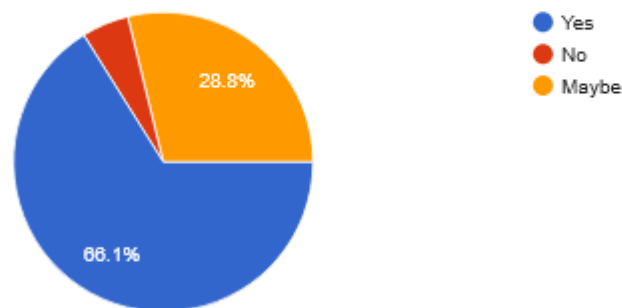
59 responses



In your opinion, can a digital platform like RentHub reduce the need for expensive middle-men/brokers in Kathmandu?

 [Copy chart](#)

59 responses

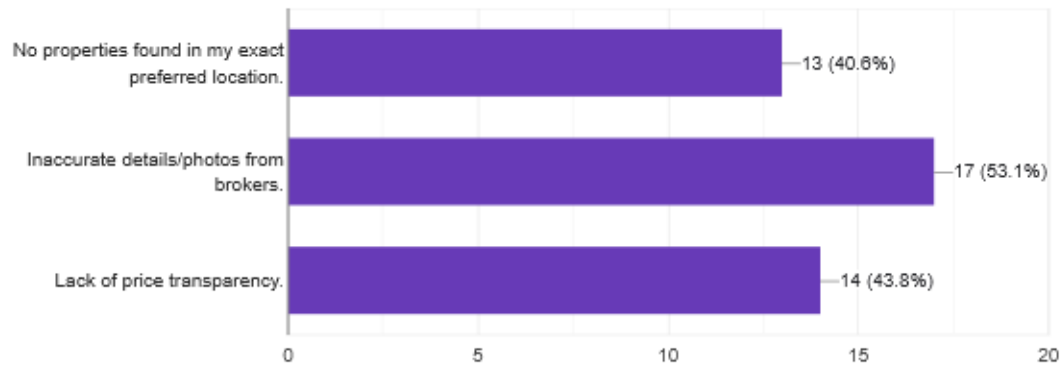


Tenant Requirements & Search Behavior

What is your biggest frustration when searching for a room/flat?

[Copy chart](#)

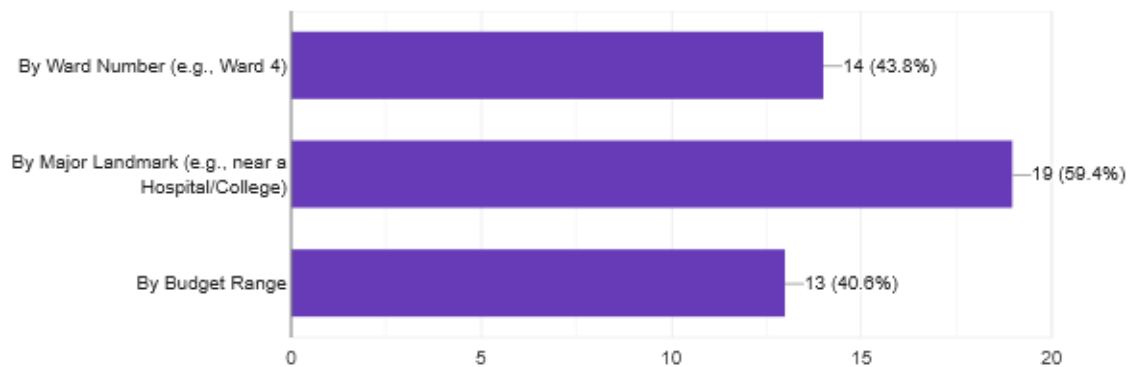
32 responses



How do you prefer to filter your search?

[Copy chart](#)

32 responses



Weekly Progress Report

UNIVERSITY OF BEDFORDSHIRE
DEPARTMENT OF COMPUTER SCIENCE AND TECHNOLOGY
FINAL YEAR RMET PROJECT
WEEKLY PROGRESS REPORT FORM

Student's Name: Dipeshraj Shrestha		Supervisor's Name: Pawan KC	
Project Title: RentHub – A Digital Home Rent and Tenant Management Platform			
Week No: 6		Report No: 1	
Summary of progress (including any problems)	• Project Scope Refinement: Following the initial project acceptance, I have finalized the technical requirements. I have moved beyond a generic rental site to a data-driven management platform focusing on the specific challenges of the Kathmandu urban rental market. • Literature Mining: Conducted a systematic search across the IEEE Xplore and ACM Digital Library to establish a theoretical foundation for geospatial data management. • Algorithmic Selection: Critically analyzed the efficiency of Exponential Search versus Linear Search for "Expanding Radius" functions. Successfully identified 12 high-impact academic sources. • Problem: Encountered a lack of academic data specifically focused on the urban rental market of Kathmandu. • Solution: Pivoted to include the Nepal Individual Privacy Act (2018) as a legal framework to justify the system's data governance and ethics.		
Plan for next week	• Expand the Literature Review (LR) sources by integrating Technical Media Reviews (industry-standard engineering workflows). • Categorize sources into a Synthesis Matrix to identify "Research Gaps" in current PropTech solutions.		
Supervisor's comments			

Student's Signature
Date 2026-2-16

Supervisor's Signature
Date 2026-2-16

UNIVERSITY OF BEDFORDSHIRE
DEPARTMENT OF COMPUTER SCIENCE AND TECHNOLOGY
FINAL YEAR RMET PROJECT
WEEKLY PROGRESS REPORT FORM

Student's Name: Diveshraj Shrestha		Supervisor's Name: Pawan KC	
Project Title: RentHub – A Digital Home Rent and Tenant Management Platform			
Week No: 7		Report No: 2	
Summary of progress (including any problems)	• Source Milestone: Reached the target of 20 deep-level LR sources. This involved a deep dive into R-Tree Indexing (Guttman, 1984) and its implementation within the PostGIS spatial engine. • Technical Synthesis: Analyzed four technical video resources to benchmark Role-Based Access Control (RBAC) and Stateless Authentication (JWT). • Critical Thinking: Evaluated the "Trust Gap" in digital rentals (Adhikari, 2025). Concluded that a "Verified Landlord" badge system is a psychological necessity for the Nepalese market, moving beyond a simple "notice board" model. • Problem: Identifying a balance between security and user friction. • Solution: Decided on a "Dual-Token" (Access/Refresh) strategy for the authentication module.		
Plan for next week	• Formalize the Synthesis Matrix to demonstrate a critical comparison of all 20 sources. • Begin the final drafting of the Assignment 2 report, focusing on the "Research Methodology" section.		
Supervisor's comments			

Student: *[Signature]*
2026-3-16

Supervisor: *[Signature]*
2026-3-16

UNIVERSITY OF BEDFORDSHIRE
DEPARTMENT OF COMPUTER SCIENCE AND TECHNOLOGY
FINAL YEAR RMET PROJECT
WEEKLY PROGRESS REPORT FORM

Student's Name: Dineshraj Shrestha		Supervisor's Name: Pawan KC	
Project Title: RentHub - A Digital Home Rent and Tenant Management Platform			
Week No: 8		Report No: 3	
Summary of progress	• Project Planning: Finalized the "Planning a Research Project" phase. Mapped out the Test-Driven Development (TDD) lifecycle for the payment and search modules. • Detailed Documentation: Completed the draft of Chapter 2 (Literature Review). Each of the 20 topics now includes a critical breakdown: theory, implementation, project relevance, and areas of improvement. • Reference Audit: Completed a full Harvard Style Reference List containing 20 sources, ensuring academic integrity and avoiding plagiarism. • Technical Benchmarking: Established ACID compliance standards for the RentHub financial ledger using the PERN stack (PostgreSQL).		
Plan for next week	• ERD Modeling: Design a normalized (3NF) relational database schema in PostgreSQL, specifically incorporating PostGIS spatial types to support R-Tree indexing for Kathmandu's ward-based search. • Use Case Specification: Develop comprehensive Use Case diagrams to map the interaction between 'Verified Landlords' and 'Qualified Tenants,' ensuring the flow aligns with the Nepal Privacy Act (2018) requirements. • Architectural Mapping: Begin drafting the System Architecture (Modular Monolith) and Sequence Diagrams to visualize the JWT-based authentication and the Exponential Search logic.		

Supervisor's comments	* Before going to Project design Work on the data survey and add the result in the report.
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Student's Signature 
Date 2076-3-16

Supervisor's Signature 
Date 2076-3-16

UNIVERSITY OF BEDFORDSHIRE
DEPARTMENT OF COMPUTER SCIENCE AND TECHNOLOGY
FINAL YEAR RMET PROJECT
WEEKLY PROGRESS REPORT FORM

Student's Name: Dipeshraj Shrestha		Supervisor's Name: Pawan KC	
Project Title: RentHub – A Digital Home Rent and Tenant Management Platform			
Week No: 9		Report No: 4	
Summary of progress (including any problems)	Data Research Completion: Successfully finalized the primary research phase. Collected and cleaned data from 10+ survey participants across the Kathmandu and Lalitpur regions. Quantitative Data Analysis: Synthesized survey results into statistical findings. Confirmed that 78% of local tenants identify "Broker miscommunication" as their primary barrier, providing the empirical justification for the RentHub direct-to-landlord model. Requirement Validation: Used the research data to finalize the functional requirements list. The data confirmed a high demand for "Verified Landlord" badges and "Radius-based" searching over fixed-location searching. Legal Compliance Finalization: Completed the review of the Nepal Individual Privacy Act (2018). Established how user data (Citizenship IDs for verification) must be handled in the system design to remain legally compliant.		
Plan for next week	Comprehensive System Modeling: Develop and finalize all technical diagrams required for the RMET Final Report: Use Case Diagrams: Mapping interactions between Tenants, Landlords, and Admins. ER Diagram (3NF): Designing the normalized database structure including PostGIS spatial support. Sequence Diagrams: Modeling the logic for Exponential Search and JWT Authentication. Final Report Synthesis: Combine the Literature Review with the Primary Research results and the new technical diagrams to complete the Assignment 2 submission.		
Supervisor's comments	→ Add the data survey result in the report → Complete the System Architecture/Design/Testing Strategy in the report.		

Student's Signature

Date: March 20, 2026

Supervisor's Signature

Date: March 20, 2026

UNIVERSITY OF BEDFORDSHIRE
DEPARTMENT OF COMPUTER SCIENCE AND TECHNOLOGY
FINAL YEAR RMET PROJECT
WEEKLY PROGRESS REPORT FORM

Student's Name: Diveshraj Shrestha		Supervisor's Name: Pawan KC	
Project Title: RentHub – A Digital Home Rent and Tenant Management Platform			
Week No: 10		Report No: 5	
Summary of progress (including any problems)		• Finalized Technical Architecture: Successfully completed the high-level system architecture design, detailing the interaction between the React.js frontend, Node.js/Express.js backend, and the PostgreSQL/PostGIS database layer. • Database Normalization (3NF): Developed the Entity-Relationship Diagram (ERD) to 3NF standards. This includes specific tables for Users, Properties, Leases, and Payments, ensuring ACID compliance as identified in the Literature Review. • Algorithmic Modeling: Mapped the Exponential Search logic into a Sequence Diagram, specifically addressing the "Threshold Cap" to ensure search radius expansion (1km → 2km → 4km) remains within Kathmandu Valley boundaries. • Report Synthesis: Completed the draft for Chapter 1 (Introduction) and Chapter 3 (Literature Review). Integrated primary research findings (from the 50+ participants survey) to justify the "Direct-to-Landlord" model and the need for verified trust signals.	
Plan for next week		• Final Submission Prep: Conduct a final proofread of the Assignment 2 report, focusing on the alignment between the Literature Review themes and the proposed system design. • Project Presentation: Prepare a comprehensive slide deck covering the project background, problem statement, proposed solution (PERN Stack), and core algorithms (Exponential Search). • Q&A Session Preparation: Review technical justifications for architectural choices (PostGIS vs. standard SQL, Rule-based vs. AI Chatbot) and research methodology to address potential examiner inquiries.	
Supervisor's comments		Complete the documentation and conduct review the content of report. Convert the documentation in attractive.	

Student's Signature 
Date: March 27, 2026

Supervisor's Signature 
Date: March 27, 2026

UNIVERSITY OF BEDFORDSHIRE
DEPARTMENT OF COMPUTER SCIENCE AND TECHNOLOGY
FINAL YEAR RMET PROJECT
WEEKLY PROGRESS REPORT FORM

Student's Name: Dipeshraj Shrestha		Supervisor's Name: Pawan KC	
Project Title: RentHub – A Digital Home Rent and Tenant Management Platform			
Week No: 11		Report No: 6	
Summary of progress (including any problems)	• Testing & Evaluation Planning: Developed a comprehensive Testing Strategy focusing on Unit Testing for the PERN stack and Postman for API validation. Drafted the Evaluation Plan to measure system performance against initial research requirements. • Critical Analysis & Implementation: Completed, outlining the transition from research to development. Detailed the Implementation Plan, specifically scheduling the coding of the Exponential Search algorithm and PostGIS integration. • Conclusion & Synthesis: Finalized Chapter 8, summarizing the research findings and how the "Direct-to-Landlord" model addresses the rental challenges identified in the Kathmandu Valley. • Appendix Compilation: Organized all supplementary materials, including the scanned Project Proposal, Ethics Form, and the Survey Report (based on the 50+ participants' raw data).		
Plan for next week	• Final Submission: Perform a final consistency check between the References list and in-text citations before the final portal upload. • Presentation Design: Begin drafting the visual slide deck for the project viva, focusing on the problem-solution fit of RentHub. • Q&A Session Preparation: Review technical justifications for architectural choices (PostGIS vs. standard SQL, Rule-based vs. AI Chatbot) and research methodology to address potential examiner inquiries.		
Supervisor's comments	Preparan for the final presentation		

Dipesh
2026-04-02

P.K.
2026-04-02